FYUGP B.A. (HONS) IN ECONOMICS/B.A.(HONS.)IN ECONOMICS

(RESEARCH) UNDER THE HONORS SCHOOL SYSTEM

(I to IV Semester)

SESSION 2024-25

Outline of Syllabi and Course Structure for FYUGP B.A.(Hons) in Economics /B.A. (Hons.) in Economics (Research) under the Honors School System for the Session 2024-25 in the Department of Economics

NEP Outlines of Syllabi and Course Structure for variousCoursesintheDepartmentofEconomics2024-25.

B.A. (Hons.) Semester I (under NEP)

Economics Core Course 1: Introductory Microeconomics (4 credits)

Economics Core Course 2: MathematicalMethodsforEconomics-I(4credits)

Skill Enhancement Course I: Data Collection and Handling Techniques (3credits) /AnyothercourseofferedbyDepartmentofMathematics/Statistics/Computer Science of the Panjab University of same or more credits

MultidisciplinaryCourse1: Fundamentals of Sociology (3credits)/any other course offered by the Panjab University of same or more credits

AbilityEnhancementCompulsoryCourse(AECC)–I(2credits)(offeredbyDepartment of English)

ValueAddedCourse1: Indian Economic Thought (2credits)/ any other course offered by the Panjab University of same or more credits

Value Added Course 2: Introduction to data Analysis Using Excel-1 (2 credits)/Any othercourseofferedbyDepartmentsofMathematics/Statistics/ComputerScience of the Panjab University of same or more credits

Minor 1

Calculus(4credits)/anyothercourseofferedbyDepartmentofMathematicsofthePanjab University of same or more credits

B.A.(Hons.) Semester II (under NEP)

EconomicsCoreCourse3: Introductory Macroeconomics–II (4 credits)

EconomicsCoreCourse4: Mathematical Methods for Economics-II (4credits)

Skill Enhancement Course 2: Data Analysis for Beginners (3 credits)/ Any other course offeredbyDepartmentsofMathematics/Statistics/ComputerScienceofthePanjabUniversity of same or more credits

MultidisciplinaryCourse2: Social Stratification (3credits)/ Any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course(AECC)–II(2credits)(offered by Department of English)

ValueAddedCourse3: Introduction to Environmental Economic**s**(2credits)/Any other course offered by the Panjab University of same or more credits

Value Added Course 4: Introduction to data Analysis Using Excel-II (2 credits)/ Any other course offered by Department of Mathematics /Statistics/Computer Science of the Panjab University of same or more credits

Minor2

Algebra (2credits)+Differential Equations (2credits)/Any other course offered by the Department of Mathematics of the Panjab University of same or more credits

B.A.(Hons.) Semester III (under NEP)

Economics Core Course 5: Intermediate Microeconomics-I (4 credits)

EconomicsCoreCourse6: Introductory Statistical Methods for Economic Analysis

(4credits) **Economics Core Course 7:** History of Economic Thought (4 credits)

Skill Enhancement Course 3:MATLAB(3credits)/Any other course offered by Department of Mathematics / Statistics / Computer Science of the Panjab University of same or more credits

MultidisciplinaryCourse3: Social Stratification (3credits)/Any other course offered by the Panjab University of same or more credits

Ability Enhancement Compulsory Course(AECC)–III(2credits)(Punjabi/Hindi) (Offered by Department of Punjabi/Hindi)

Minor3

Real Analysis (4credits)/Any other course offered by Department of Mathematics of the Panjab University of same or more credits

B.A.(Hons.)Semester-IV(under NEP)

EconomicsCoreCourse8: Intermediate Microeconomics-II(4credits)

EconomicsCoreCourse9: Intermediate Macroeconomics(4 credits)

EconomicsCoreCourse10: Introductory Econometrics(4 credits)

EconomicsCoreCourse11: Growth Models(4 credits)

Ability Enhancement Compulsory Course(AECC)–III (2credits) (Punjabi/Hindi) (Offered by Punjabi/Hindi Department)

Minor4

Statistics (4credits)/ Any other course offered by Departments of Statistics of the Panjab University of same or more credits

Course Structure and Syllabi for FYUGPB.A. (Hons.) Economics under the Honours School System for 1st,2nd, 3rd and 4thSemester for 2024-25 Session

Course Structure for FYUGP B.A. (Hons.) Economics under the Honours School System for 1st, 2nd, 3rd and 4th Semester for the Session 2024-25

Semester-I			
Course	**Course Code**	**Paper**	**Credits**
EconomicsCoreCourse1	ECODSC 1	Introductory Microeconomics	4
EconomicsCoreCourse2	ECODSC 2	Mathematical Methods for Economics-I	4
SkillEnhancementCourse1	ECO SEC 1	Data Collection and Handling Techniques	3
Multi disciplinary Course 1	MDSC1	Fundamentals of Sociology	3
Ability Enhancement CompulsoryCourse1	AECC1	English	2
ValueAddedCourse1	*ECOVAC1	Indian Economic Thought	2
ValueAddedCourse2	ECOVAC2	Introduction to data Analysis Using Excel-I	2
Minor:			
MathematicsCourse1:	**MAT-DSC-101	Calculus	4
Total Credits			24
Semester-II			
EconomicsCoreCourse3	ECODSC 3	Introductory Macroeconomics	4
Economics Core Course 4	ECODSC 4	Mathematical Methods for Economics-II	4
SkillEnhancementCourse2	ECO SEC 2	Data Analysis for Beginners	3
Multi disciplinary Course 2	PUBMDSC 2	Basics of Indian Administration	3
Ability Enhancement CompulsoryCourse2	AECC2	Course offered by Department of English	2
ValueAddedCourse3	*ECOVAC3	Introduction to Environmental Economics	2

ValueAddedCourse4	ECOVAC4	Introduction to data Analysis Using Excel-II	2
Minor:			
MathematicsCourse2	MAT-DSC151	Algebra and Ordinary Differential Equations	4
Total Credits			24
Semester-III			
EconomicsCoreCourse5	ECODSC 5	Intermediate Microeconomics-I	4
Economics Core Course 6	ECODSC 6	Introductory Statistical Methods for Economic Analysis	4
Economics Core Course 7	ECODSC 7	History of Economic Thought	4
SkillEnhancementCourse3	ECO SEC 3	MATLAB	3
Multi disciplinary Course 3	MDSC3	Social Stratification	3
Ability Enhancement Compulsory Course3	AECC3	Course offered by Department of Punjabi/Hindi	2
Minor:			
MathematicsCourse3	MAT-DSC-251	Real Analysis	4
Total Credits			24
Semester-IV			
EconomicsCoreCourse8	ECODSC 8	Intermediate Microeconomics-II	4
Economics Core Course 9	ECODSC 9	Intermediate Macroeconomics	4
Economics Core Course 10	ECODSC 10	Introductory Econometrics	4
Economics Core Course 11	ECODSC 11	Growth Models	4
Ability Enhancement CompulsoryCourse4	ECOAECC4	Course offered by Department of Punjabi/Hindi	2
Minor:			
StatisticsCourse1	STAT-M-111(Theory)+ STAT-M-112(Practical)	Statistical Methods	4
		Internship/CW/FP	2
Total Credits			24

*** ECODSC1(4credits) +ECOVAC1(2credits)would be offered as Minor-1 (6 credits) to Non Economics Honors students in the Semester-I**

*** ECODSC3(4credits)+ECOVAC3(2credits) would be offered as Minor-2(6 credits) to Non Economics Honors students in the semester-II**

Note:

****1. A 6 credit minor(150marks)would be treated at par with a 4 credit minor (100 marks) and 150 marks to be normalized to 100 for BA. Thus, Maths minor students would have more than 4 teaching hours every week.**

***2. Regarding balancing of B. Sc BA Minor credits the first year A B.Sc. student who opts for a minor of 4 credits from B.A. will cover the remaining 2 Credits from the VAC of the same subject of 2 credits thus earning total 6 credits in that minor and would have two VAC courses in the first and second semester. In the first semester 1 to 2 VAC courses are allowed.**

Syllabi of B.A.(Hons.) Semester-I (under NEP)

ECODSC1: INTRODUCTORY MICRO ECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L=60hours(4 hours aweekfor15weeks=01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective:-This course is designed to expose the students to the basic principles of microeconomic theory. The emphasis will be on thinking like an economist and the course will illustrate how microeconomic concepts can be applied toanalyze real-life situations.

Course Outcomes: -1.To train them to think like an economist.

2.To rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, in general, a static and partial equilibrium framework.

Pedagogy of the Course Outcomes:-80%Classroomteaching.
20%InternalAssessment.

Instructions for Paper-setter and candidates:-

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT-I

Exploring the subject matter of Economics. Why study economics? Scope and method of economics; the economic problem: scarcity and choice; the question of what to produce, how to produce and how to distribute output

Cardinal Utility Analysis.

Ordinal Utility Analysis: The consumption decision - budget constraint, consumption and income/price changes, demand for all other goods and price changes; description of preferences(representing preferences with indifference curves); properties of indifference curves; consumers optimum choice; income and substitution effects.

UNIT-II

Theory of production: laws governing short run and long run. Producers equilibrium and expansion path

Cost Theory: Short run and long run cost curves.

The Firm and Industry under Perfect Market Structure: Price and out put determination; short run and long run.

UNIT-III

Determinants of individual demand/supply; demand/supply schedule and demand/supply curve; market versus individual demand/supply; shifts in the demand/supply curve, demand and supply together; how prices allocate resources. The basic competitive model. Supply and Demand How Markets Work, Markets and Welfare Markets and competition Elasticity of demand: its measurements, applications and determinants .Consumer surplus; producer surplus and the efficiency of the markets.

Concept of dead weight loss; Govt. intervention in the form of Taxes and Subsidies.

UNIT-IV

Single seller monopoly: Price and out put determination

Input Markets Labour and land markets - basic concepts (derived demand, productivity of an input, marginal productivity of labour, margin alrevenue product); demand for labour; input demand curves;

shifts in input demand curves; competitive labour markets and firms profit maximization conditions in input market ; and labour markets and public policy. Derivation of labour supply and savings decision -choice between leisure and consumption.

Essential Readings:-

1. N. Gregory Mankiw. (2007) *Economics: Principles and Applications*. (4th ed.). India edition by South Western, a part of Cengage Learning, Cengage Learning India Private Limited.
2. Karl, E. Case., and Ray C. Fair, (2007). *Principles of Economics (*8th ed.).Pearson Education Inc.

Suggested Readings:-

1. Joseph E. Stiglitz, and Carl E. Walsh. (2007) .*Economics* (4th ed.). W.W.Norton & Company, Inc., New York, International Student Edition.
2. Salvotore. D(2006)*Theory and Problems of Microeconomic Theory* (Sachems series)(3rd ed.)Tata McGraw-Hill Publishing Company Ltd.
3. Salvotore. D. (2007). *Micro economic theory and Applications* (4th ed.)Oxford University Press.

ECODSC2: MATHEMATICAL METHODS FOR ECONOMICS–I

Topology of the code	**Major**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorial, Practical)	**L=60 hours(4hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+Internal Assessment20marks**
Time	**3hours**

Course Objective:-This is the first part of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in thissyllabus.Inthiscourse,particulareconomicmodelsarenottheends,but the means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication with which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes: - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2.Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration of functions and their simple economic applications. 4. Students are able tounderstand and apply the Difference Equations and their simple economic applications.

Pedagogy of the Course Out comes:-80%Classroomteaching.

20%InternalAssessment.

Instructions for Paper-setter and candidates:

The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

There shall be **9** questions in all.

The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Preliminaries: Essentials of Set Theory, Logic, Mathematical Proofs, Mathematical Induction. The real Number System.

Functions of one real variable : Graphs of functions, Linear, quadratic, polynomial, power, exponential, logarithmic.

Properties of Functions : Shifting of Graphs, New functions from Old, Inverse functions, Graphs of Equations, Distance in Plane.

Limits, continuity and series.

UNIT-II

Differentiation: Differentiation concepts, properties with respect to various operations and applications; second and higher order derivatives: properties and applications.

Convex and Concave Functions and Optimization:

Convex and Concave functions :Properties and applications; Local Extreme Points (Maxima and Minima) and applications.

UNIT-III

Integration: Indefinite Integrals, Definite Integrals, Methods of Integration by Parts, Substitution and partial fractions and simple economic applications

UNIT-IV

Difference equations : Introduction, solution of difference equations up to2^{nd} order, simple economic applications.

Essential Readings:-

K. Sydsaeter and P. Hammond, (2016). *Mathematics for Economic Analysis (11th Edition), Pearson:NewDelhi{Chapter1:Section1.4,Chapter6:Section6.1to6.7,Chapter9: section9.1to9.6;Chapter10:section10.1to10.4;Chapter11:Section11.1to11.4; Chapter20:section20.1to20.5}*

K. Sydsaeter, P. Hammond, A. Strom and A.Carvajal (2021).Essential Mathematics for EconomicAnalysis(6thedition),Pearson.{Chapter1:Section1.1to1.4;Chapter4: section4.1to4.10;Chapter5:Section5.1to5.5;Chapter6:section6.1to6.11;Chapter8: Section8.1to8.6;chapter9:Section9.1to9.6;Chapter10:section10.1to10.7}

Suggested Readings:-

Weber,JeanE(1982).Mathematical*AnalysisBusinessandEconomicApplications* (4thEd.)Yamane Taro(1972).Mathematics for Economists.

ECOSEC1: Data Collection and Handling Techniques

Topology of the code	**Skill Enhancement Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures,Tutorials, Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=0, P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60marks+InternalAssessment 15 marks**
Time	**3hours**

Course Objective:- To provide skills to undertake data based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes :-It will enable the students to qualify forjobs relating to fieldinvestigation in any field of social sciences.

PedagogyoftheCourseWork:-80%Classroomteaching.
20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

- Themaximummarksforthepaperwillbe75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question a per in this regard.

- There shall be **7**questionsinall.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of 3marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT-I

Research: Meaning, objectives, Characteristics, Types, Process and Utility, Research method Vs. Methodology. Research Design-Meaning, Types and Features

of Good Research Design. Types of Data: Primary vs Secondary, Cross Sectionor Micro Data, Time Series Data, Pooling of data, Panel data, Censusvs sampled data.

UNIT-II

Sampling – Concepts of Statistical Population, Sample, Sampling Frame, Sampling and non-Sampling Errors, Sample Size. Census and Sampling survey. Characteristics of good sampling.
Sampling Techniques: Simple Random Sampling; PPS Sampling; Stratified Random Sampling; Multistage Sampling (Applications only), determination of sample size.

UNIT-III

Methods of collection of Primary data (observation method, interview method, collectionofdatathroughquestionnaires- Designing ofquestionnaire, collectionof data through schedules- Designing of schedules).Measurement and scaling, validity and reliability.

CollectionofSecondaryData
Selectionofappropriatemethodofdatacollection.

**** Practical Work: Students are required to formulate a research problem and need to prepare a questionnaire for the same along with appropriate measurement and scaling techniques.

EssentialReadings:-

1. Kothari,C.R.(2009).*ResearchMethodsandTechniques*, WileyEastern.

SuggestedReadings:-

1. Bethlehem,J.(2009).*Appliedsurveymethods:Astatisticalperspective.*Wiley.
2. Cochran,W.(2008).*Samplingtechniques,3rded.*Wiley.
3. Cooper,D.,Schindler, P.,Sharma, J.(2012).*Businessresearchmethods*,*12thed.* McGraw-Hill.
4. Flick,U.(2012).*Introducingresearchmethodology:Abeginner'sguide to doing aresearch project*. Sage Publications.
5. Groves,R.,Fowler,F.,Couper,M.,Lepkowski,J.,Singer,E.,Tourangeau,R. (2009).
SurveyMethodology.Wiley.
6. Kumar,R.(2014).*Researchmethodology:Astepbystep guide forbeginners*, *4th ed.* Sage Publications.

MDSC1:FUNDAMENTALS OF SOCIOLOGY

Topologyofthecode	**MultidisciplinaryCourse**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours(3hoursaweekfor15 weeks= 01semester), T=0, P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60marks+Internal Assessment15 marks**
Time	**3hours**

Course Objective: - The systematic study of human behaviour and human society is a relatively recent development. The central emphasis of the paper would be to develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to development of a sociologicaloutlook. The course would also discuss the works of founding father which led to the development to sociological thinking.

CourseOutcomes:-1.Studentsareabletostudyhumanbehaviorand human society relative to recent development. 2. Students develop aconceptualclarityregardingthebasicsociologicaltermsandthemes,thus leading to a development sociological outlook.

PedagogyoftheCourseWork:- 80%Classroomteaching. 20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment will be of15 marks.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**7**questionsinall.

The first question is**compulsory**comprising 8 short answer type questions spread over the whole syllabus. The candidates are requiredto attempt 5 questions. Each question shall be of 3marks (5x3=15).

Restofthepapershallcontainthreeunits.Two questionsshallbeasked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each questionwill carry15 marks. (15x3=45).

UNIT– I

Developing Sociological Outlook*:* Definition and Scope; Significance and Usesof Sociology; Relationship of Sociology with other Social

Sciences – Anthropology, Economics, Political Science, Psychology, Philosophy and History.

UNIT– II

Basic Concepts: **Society** – Meaning and Characteristics, Theories of Origin, Individual and Society. Community, Association and Institution – Meaning, Characteristics and Differences.Groups – Definition, Characteristics andClassification.

UNIT– III

Culture and Socialization: Culture-Definition, Cultural Lag, Acculturation,Assimilation, Pluralism. Culture and Personality. ***Civilization:***Definition,TraditionalandModern.Socialization–Meaning, Stages,Agencies.

EssentialReadings:-

1. Giddens,Anthony.*Sociology, Cambridge(*4th ed.).PolityPress.

SuggestedReadings:-

2. Haralambos,Michael.(2015).*Sociology:ThemesandPerspectives*.Oxford.
3. Davis,Kingley.(2002).*HumanSociety.*Surjeet Publications,Delhi.
4. Maclver,RS&PageCharles.*Society:AnIntroductory Analysis* (Latested.). Macmillan India Ltd.
5. Johnson,Harry(2003).*Sociology:ASystematicIntroduction*.Allied Publishers,NewDelhi.
6. InkelesAlex.(2004).*WhatisSociology?AnIntroductiontothe Disciplineand Profession*. Prentice-Hall, India.
7. Bradshaw,YorkW,et.al.(2001).*SociologyforaNew Century*, PinerForge Press, California.
8. Johnson,DoleyPaul.(1986).*SociologicalTheory*(Orlater ed.). NewYork,McMillan.
9. Ritzer,George.(2011).*SociologicalTheory.*London(Orlatested.).McGraw-Hill.
10. CoserLewis.(2004).*MastersofSociologicalThought*,Rawat Publications,NewDelhi.
11. Rao Shankar C.N. (2012).*Sociology*: *principles with anIntroductionto Social Thought*.S.Chand and Company Pvt. Ltd.

ECOVAC-1:INDIAN ECONOMIC THOUGHT

Topologyofthecode	**Value Added Course**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=30hours(2hoursaweekfor15weeks= 01semester), T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment 10 marks**
Time	**2hours**

CourseObjective:-Thecourseaimstoenablethestudentstoacquireand demonstrate the knowledge on economic thoughtwith its historical perspective.

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Course Outcomes : - 1. Studentsthroughthis course become familiar withthe historicalevolution ofeconomic ideas. Theyare able to learn the foundational contributions of the classical economists to the economic literature.

2. Students are able to develop understanding about the philosophical foundations of economic analysis of capitalist system. They also learn the theories of imperialism which are important to understand the exploitation process ofthird world countries by the metropolis nations.

3. Studentsareabletotheoriginofmarginalist,neoclassicalandAustrianschool of thought and their main features. This is important to understand the differencesandsimilaritiesamongtheir ideas.4.Studentsareabletolearnabout the Keynesian revolutionin economic theory. They also understand the causesofgreat depression. Moreover, they learn about the Kaleckian version ofmacro economics.

PedagogyoftheCourseWork:-80%Classroomteaching.
20%InternalAssessment.

Instructions for Paper-setter and candidates:

• The maximummarksforthepaperwillbe 50. Thequestionpaperwillbeof 45 marks andcontinuous evaluation 05 marks. Time allowed will be 1.5 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

• Thereshallbe5questionsinall.

The first questioncompulsorycomprising 5 short answer type questions spread overthewholesyllabus.Thecandidatesarerequiredtoattempt 3questions.Each question will carry 5 marks (5x3= 15).

Rest ofthe paper shall contain two units. Two questions shall be asked fromeach unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattempt one

question fromeachunit. Each question will carry 15 marks (15x2=30).

UNIT-I

Pre-independenceIndianEconomicthought

Kautilya'sArthasastra:Theeconomic functionsofthestate,Taxationandpricing policy, Theland system

DadabhaiNaorojiandthedraintheory:What isthedrain?,CriticismsandLong-run relevance

Ranadeandtheeconomicsofdevelopment:Povertyand industrialisation, Agrarian policy,Railway investment.

UNIT-II

IndianEconomicthoughttowardsindependence

Gokhaleand theeconomicsofeducation

Gandhian economics: Methodological issues, Consumption behaviour: the limitationofwants,Swadeshi;Trusteeshipandindustrialrelations;Charity, leisure and the sanctity ofwork

EssentialReadings:-

1. Dasgupta,A.K.(1993).AHistoryofIndianEconomicThought (1sted.). Routledge.

SuggestedReadings:-

2. Kautilya.&Shamasastry,R.(1967).*Kautilya'sArthasastra*.Mysore,India :MysorePublishingandPrintingHouse.
3. Naoroji,D.(1901).PovertyandUn-BritishRuleinIndia.United Kingdom: S.Sonnenschein.
4. Chandra,B.,Ranade,M.G.(1990).Ranade'sEconomicWritings. India: GianPublishing House.
5. Gokhale, G. K. (2018). Gopal Krishna Gokhale: Select SpeechesandWritings.India:NationalBookTrust,India.
6. VillageSwarajbyM.K.Gandhi.(2021).(n.p.):PrabhatPrakashan.
7. Dasgupta,A.K.(1996). Gandhi'sEconomicThought.(n.p.):Taylor &Francis.

ECOVAC-2: INTRODUCTION TO DATA ANALYSIS USING EXCEL-I

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=30hours(2hoursaweekfor15 weeks= 01semester),T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+Internal Assessment10 marks**
Time	**2hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoninginvolved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjectivetype questions/ answers (bothshort and essaytype). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work: The course relies on a combination of lectures, solvingproblems anddiscussingof academicarticlesorreal-lifesituations. Teacher will assign topic for assignmentson contemporarythemes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacherinteractions.

Instructions for Paper-setter and candidates:

• The maximummarksforthepaperwillbe 50. Thequestionpaperwillbeof 45 marks andcontinuous evaluation 05 marks. Time allowed will be 1.5 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

• Thereshallbe5questionsinall.

The first questioncompulsorycomprising 5 short answer type questions spread overthewholesyllabus.Thecandidatesarerequiredtoattempt 3questions.Each question will carry 5 marks (5x3= 15).

Rest ofthe paper shall contain two units. Two questions shall be asked fromeach unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattempt one question fromeachunit. Each question will carry 15 marks (15x2=30).

CourseOutcomes

After successful completion, the students will be able to understand the use ofMS-

Excel in data handling and presenting. The filtering techniques, lookup methods, pivot tabling, graphics presentation, matrix analysis etc. shall sharpen a student in the field of quantitativeeconomics and econometrics.

Unit-I

IntroductiontoSpreadsheets

Introductiontospreadsheets,readingdata,manipulatingdata.Arithmeticoperationsand basic functions.

Spreadsheet FunctionstoOrganizeData
IntroductiontosomemoreusefulfunctionssuchastheIF,nestedIF,VLOOKUPand HLOOKUP functions in Excel.

Unit-II

IntroductiontoFiltering,PivotTables and Charts

Introduction to the Data filtering capabilities of Excel, the construction of Pivot Tables to organize data and introduction to charts in Excel. Constructing various Line, Bar and Pie charts. Using the Pivot chart features ofExcel. Understanding and constructing Histograms and Scatterplots.

References:

Lloyd,Chris, J. (2011).DataDrivenBusinessDecisions, JohnWiley&Sons.

Levine,D.M.,Stephen,D.andSzabat,K.A.(2016).StatisticsforManagersUsing Microsoft Excel, Pearson Education.

Day,AlastairL.(2013).MasteringFinancialMathematicsinMicrosoftExcel,Prentice Hall.

Piascik,Chester(2001).AppliedMathematicswithMicrosoftExcel,PennsylvaniaState University.

Jelen,Bill(2013). Excel2013:ChartsandGrapMicrosoftExcel.QuePublishing

MAT-DSC-101: Calculus

Credits:6(L=5;T=1;P=0)

TotalMarks:150(IncludingInternalAssessment=30) **TimeAllowedforExamination:**3hrs.

InstructionsfortheCandidatesandPaperSetters

Candidateswillbeaskedtoattemptfivequestionsoutofnine,carryingequalmarks.QuestionNo.1 spreadoverthewholesyllabuswillbecompulsory.

Therewillbetwoquestionsfromeachunitandthestudentswillhavetoattemptonefromeachunit.

Course Outcome: The main goal of this course is to deliver the basics of sequences andseries;differentialandintegralcalculus,forrealaswellasmultivariatefunctions.It isexpectedthatthestudentsdevelopatasteofwritingproofs, particularlyforUnitIand Unit III, rather than applying formulas only.

Unit-I

Differential Calculus: Precise definition of limit, continuity, one-sided limit, limits involving infinity,asymp- totes of graphs, tangents and the derivative at a point,the derivative of a function,extreme values of functions, mean value theorem, monotone functions and the first derivative test, test for concavity, tracing of curves.

(Scope:Sections2.3–2.6,3.1,3.2,4.1–4.4of(A)).

Unit-II

Integral Calculus: Riemann sums, definite integrals, area between curves, volumes using cross sections and cylindrical shells, arc length and areas of surfaces of revolution.

(Scope:Sections5.1,5.6,6.1–6.4of(A)).

Unit-III

Infinite sequences andSeries:sequences, infinite series, the integral test, comparison test, the root and ratio test,alternatingseries,absoluteandconditional convergence, power series, Taylor and Maclaurin series, Power series, convergence of Taylor series,application of Taylor series.

(Scope:Sections10.1-10.10of(A).)

Unit-IV

Polar Coordinates: Polar Coordinates, graphing in Polar coordinates, areas and lengths in Polar coordinates.

Multivariable Functions: Limits and continuity for functions of several variables, partial derivatives, the chainrule, directional derivatives, gradient vectors, tangent planes, extreme values and saddle points. (Scope:Sections 11.3-11.5, 14.2 – 14.7 of (A)).

EssentialTextbooks

G.B.THOMAS,M.D.WEIRANDJ.R.HASS,Inc.,Thomas'Calculus,(12th ed.), *Pearson*Education,2014.

FurtherReadings

1. J.L.TAYLOR*FoundationsofAnalysis,*AmericanMathematicalSociety,2012.
2. S.NARAYAN,*IntegralCalculus,*S.ChandandCompanyLtd,2001.
3. M.J.STRAUSS,G.L.BRADLEYANDK.J.SMITH,*Calculus,*(3rd ed.), Pearson Education, 2007.
4. H.ANTON,I.BIVENSANDS.DAVIS,*Calculus,*(7th ed.),JohnWileyandSons,2002.
5. R.COURANTANDF.JOHN,*IntroductiontoCalculusandAnalysis(VolumesI&II),* Springer-Verlag,1989.

AECC1: ABILITY ENHANCEMENT COURSE

CourseTitle: AEC–I:LanguageSkills:Listeningand Speaking
CourseCode: ENG(AEC)—030
TypologyofCourse: AEC
CourseDuration: 60hoursofpracticalinonesemester(4hr/week)
Credits: 2credits
MaximumMarks: 50marks(30Theory+15Practical+5InternalAssessment)

LearningObjectivesoftheCourse:

- To understand the structure of language and how language varies over time, acrosssocial situations and social groups
- To be proficientin articulating thoughts and communicatingideasin English language
- To participate in debates and discussions where ideas are tested and sharpened andhoning the skill of public speaking
- Studentsshouldbeabletohold meaningfulconversations
- Studentshould beabletoeffectivelyuseEnglishlanguagetomeet professionaland personal goals

PedagogyoftheCourse:

a) Thepaperrequires4hoursofpracticalsperweekduringwhichstudentswillbetaught and explained the theory of communication through practical and hands-on training in various aspects of communication.

b) Thepracticalevaluation,whichcarries15 marks,willbedone byanexternalexaminer either from the same institution/department or from outside.

CourseContent:

UnitI: SoundsofEnglish(**Text**:*EnglishLanguageSkills:APracticalApproach*, Cambridge).

Unit2: LanguageUseinVariousSituations(**Text**: *EnglishLanguageSkills:APractical Approach*, Cambridge).

Unit3: ArtofPublicSpeaking(**Text**:*CommunicationSkills*bySanjayKumar&Pushp Lata. OUP 2023)
a. GroupDiscussion
b. Interview

SuggestedReadings:

- *FluencyinEnglishPartII*, OxfordUniversityPress.
- *ATextbookofPhoneticsforIndianStudents*,T.Balasubramian,Macmillan.

BusinessEnglish,Pearson
- *Language,LiteratureandCreativity*,OrientBlackSwan.
- *LanguagethroughLiterature*,GauriMishra,RanjanaKaul,BratiBiswas(Eds), Primus Books.
- *EnglishCommunication,*MacmillanEducation.
- *EnrichYourEnglish*,S.R.InthiraandV. Saraswathi,OxfordUniversityPress.
- *CommunicationandLanguageSkills*,SanghitaSen,et al,CambridgeUniversity Press.
- *EnglishandCommunicationSkillsI*,TasneemShahnaaz, C.SavithaandCherlyl Rao, Cambridge University Press.
- *PracticalEnglish*,K.K.Singh,APHPublishingCorporation.

Testing Scheme:

Max.Marks: 50 marks

Theory:30marks

Practical:15 marksInternal Assessment:05marks

Time:1.5Hours

Note-Questionno.1iscompulsory.AttemptanyfourquestionsfromQ2to7.

Shortnotes.**Twooutoffive**tobeattempted innot morethan150wordseach. (10marks)

Q2.to Q.7. fromthe prescribed syllabus ofListening and Speaking. **Fourout of six**questions tobe attempted in not more than 150 words each.
(20marks)

Syllabi of B.A.(Hons.) Semester-II (under NEP)

ECODSC3: INTRODUCTORY MACRO ECONOMICS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60hours(4 hours aweekfor15weeks=01semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Course Objective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "IntroductoryMacroeconomics" equips the students at the undergraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

PedagogyoftheCourseWork

Thecoursereliesonacombinationoflectures, solvingproblems, anddiscussingofacademic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

After completion of the course, a student must be capable of understanding various measures of National income and accounting. Also the paper shall inculcate the fundamental macroeconomic modelling abilities among students to handle macro-economic relationships for their empirical testing. After completion of course, the candidate shall be specifically able to identify the determinants of consumption and investment functions along with the measuresof national income;a major cause of variation in two functions.

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.
2. Thereshallbe9questionsinall.
3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each).It would carry15 short questions, spread over the entire syllabus. The candidate will be requiredtoattempt any10short answertypequestions.Eachshort answertypequestionwould carry 2 marks ($10 \times 2 = 20$).
4. RestofthepapershallcontainÂ4units. Eachunit shallhavetwoquestionsandthecandidatesshall berequired to attempt onequestion fromeach Unit— 4 inall. Eachquestionshallcarry15 marks ($15 \times 4 = 60$).

Unit-I
(MeasurementofNationalIncome)

Conceptual Introduction: Aggregates of National and Personal income. Measure of disposable income. The measurement of these aggregates at market and constant prices.

Methods of National Accounting: Expenditure, Value Added, and Income Methods. Mixed methods of national accounting.

Social Accounting: Techniques of Maintaining National Accounts such as Double Entry system, Social Accounting Matrix, National Accounting Equations. The Input Output Tables as base of Social Accounting Matrix. Introduction to Supply and Use Tables.

Unit-II

(FundamentalTheoriesofIncomeandEmployment Determination)

***IncomeandEmploymentDetermination*:**IntegratedClassicaland KeynesianModelsofIncomeand Employment Determination; commodity, money (including bond market of Keynes), and labour markets.

Wage-PriceFlexibilityandAutomaticFullEmployment:ClassicalVersus KeynesianApproach.

Unit-III

(ConsumptionTheories)

Consumption and Consumption Function: Keynes Consumption and saving functions under Psychological law of consumption, Consumption Puzzle: Absolute Income hypothesis, Relative Income hypothesis, Permanent Income hypothesis and Life Cycle Hypothesis.

ConsumptionunderUncertainty: RandomWalkHypothesis;Interest RateandSaving; Consumption and Risky Asset: Consumption CAPM.

UNIT-IV

(InvestmentTheories)

Investment and Investment Function: Type of Investment, Role of investment using Investment Multiplier, Classicaland KeynesianTheories ofInvestment, Accelerator TheoryofInvestment, Neo-Classical Theory of Investment and Tobin's-q Theory of Investment. Effects of Uncertainty, Kinked and Fixed Adjustment Costs, Investment in the Housing Market.

RecommendedReadings:

Ackley,G.(2007).MacroeconomicTheory***. Macmillan.***

Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment***. Palgrave Macmillan.***Beckerman,W.(1980).IntroductiontoNationalIncome Analysis, *LittlehamptonBookServicesLtd*. ***Blanchard, O. (2009).*** Macroeconomics***. New Delhi: Pearson Education Inc.***

Blanchard, O.J.&Fischer. (1996). LecturesonMacroeconomics***, Delhi: PHILearningPvt.Ltd.***

Branson, W. H. (1989). Macro Economic Theory and Policy ***(3rd ed.). Harper Collins.***

Cooper,R.,Edey, HaroldC., andPeacock,A.T.(2003).NationalIncomeandSocialAccounting. *Routledge.*

D'Souza,E.(2008).Macroeconomics. ***PearsonEducation.***

Dornbusch,R.,Fisher,S&Startz,R.(2002).Macroeconomics***,(9thed.).NewDelhi:Tata McGraw Hills.***

Levacic,R.&Rebmann,A.(1982). Macroeconomics:AnIntroductionto KeynesianNeoclassical Controversies***. Macmillan.***

Miller,R.L.&Pulsinelli,R.(1986).Macroeconomics.***NewYork:Harper&Row. Ola Olsson (2010).*** Essentials of Advanced Macroeconomic Theory***, Routledge.***

Romer,D.(1996).AdvancedMacroeconomics***.McGraw-Hill.***

***UmaDattaRoyChoudhury*(2000).**NationalIncomeAccounting,*Macmillan*.

FurtherReadings:

Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. ***New Delhi: Pearson Education Inc.***

Dornbusch,R.(1980).OpenEconomyMacroeconomics***.NewYork:WeltwirtschaftlichesArchive.***

Frisch, H. (1983). Theories of Inflation***. Cambridge: Cambridge University Press.***

Froyen,R.T.(2002).MacroEconomics. ***NewDelhi:PearsonEducationInc. Gordon, R. J. (2012).*** Macroeconomics***. Delhi: PHI Learning Pvt. Ltd.***

Heijdra,B.J.(2017).TheFoundationsofModernMacroeconomics***.OxfordUniversityPress. Hicks, J. R. (1950).*** A Contribution to the Theory of Trade Cycle ***Oxford*:** Clarendon Press**.**

Lucas,R.(1981).StudiesinBusinessCycleTheory***,Massachusetts:MITPressCambridge. Makin, A. J. (2002).*** International Macroeconomics***. Princeton Hall.***

Mankiw,N. G.&Romer,D.(1991).NewKeynesianEconomics.MIT,2Volumes.

Mathews, R. C. O. (1964). The Trade Cycle. ***Cambridge: Cambridge University Press.***

Pentecost,E.(2000). Macroeconomics–AnOpenEconomyApproach. ***MacMillanPressLtd.***

Raghbendra,J.(1964). ContemporaryMacroeconomics:TheoryandPolicy. ***NewAgeInternationalPublishers.***

Sherffrin,S.(1999).RationalExpectations.***CambridgeUniversityPress.***

Snowden,B&Vane,H.R.(2005). ModernMacroeconomics:ItsOrigin,Development andCurrent Stat., ***UK: Edward Press.***

Wendy,Carlin/Soskice,D.(2007).Macroeconomics:Imperfections,InstitutionsandPolicies***. Oxford University Press.***

ECODSC 4: MATHEMATICALMETHODSFOR ECONOMICS–II

Topologyofthecode	**Major**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures, Tutorials,Practical)	**L=60hours(4hoursaweekfor15weeks=01semester),T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20marks**
Time	**3hours**

Course Objective :-This is the first of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication atwhich the material is to be taught is indicated bythe contents of the prescribed textbook.

Course Outcomes : - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3.Studentsareabletounderstandand applytheintegrationo functionsandtheir simple economic applications. 4. Students are able to understand and apply the DifferenceEquations and their simple economic applications.

PedagogyoftheCourseOutcomes:-80%Classroomteaching.
20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

- Themaximum marks for the paper will be 100.The question paper will be of 80marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**9**questionsinall.

Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadover thewholesyllabus. Thecandidatesarerequiredto attempt 10questions. Eachquestion shall be of **two** marks (10 x 2= 20).

Rest ofthe paper shallcontain four units. Two questions shall be asked fromeachunit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Differential Equation*:* Introduction; Solution: Variable Separable Case, Homogeneous Case, Standard Linear Differential Equation, Bernoulli's form, Exact Equation; Solution of Linear differential Equation with Constant Coefficients; Simple Application Questions.

UNIT-II

Linear algebra: Vector spaces: algebraic and geometric properties, scalar products, norms, orthogonality; linear transformations: properties, matrix representations and elementary operations; systems of linear equations: properties of their solution sets; determinants: characterization, properties and economic applications.

UNIT-III

Functions of several real variables: Geometric representations: graphs and level curves; differentiable functions: characterizations, properties with respect to various operations and applications; second order derivatives: properties and applications; the implicit function theorem, and application to comparative statics problems; homogeneous and homothetic functions: characterizations and economic applications.

UNIT-IV

Multi-variable optimization: Convex sets; geometric properties of functions: convex functions, their characterizations, properties and applications; further geometric properties of functions: quasi convex functions, their characterizations, properties and applications; unconstrained optimization: geometric characterizations,characterizationsusingcalculusandapplications;constrainedoptimizatio nwith

equality constraints: geometric characterizations, lagrange characterization using calculus and applications; properties of value function: envelope theoremand economic applications.

EssentialReadings:-

K.SydsaeterandP.Hammond(2002).*MathematicsforEconomicAnalysis*,

Pearson Educational Asia Delhi.

K. Sydsaeter and P. Hammond, (2016). *MathematicsforEconomic Analysis (11th Edition), Pearson: NewDelhi*

{Chapter5,Chapter12,Chapter14:Section14.1-14.3,Chapter15:Section15.1-15.3,Chapter16, Chapter17,Chapter18:Section18.1-18.7,Chapter21}

SuggestedReadings:-

Weber,JeanE.(1982*).MathematicalAnalysisBusinessandEconomicApplications*(4thEd.)
Yamane,Taro (1972). *Mathematics for Economists*.

ECOSEC2: DATA ANALYSIS FOR BEGINNERS

Topologyofthecode	**SkillEnhancementCourse**
LeveloftheCourse	**100-199**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=45hours(3 hours aweekfor15weeks=01semester), T=0, P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60marks+InternalAssessment15 marks**
Time	**3hours**

CourseObjective:-Toprovideskillsto undertakedatabasedresearchsothatstudentsgetthe competencyof executing sample surveys and knowledge about secondarydata collection.

CourseOutcomes: -It willenablethestudentstoqualified for jobsrelatingto field investigation in any field of social sciences.

PedagogyoftheCourseWork:-80%Classroomteaching.

20%InternalAssessment.

InstructionsforPaper-setterandcandidates:

- Themaximummarksforthepaperwillbe75. Thequestionpaperwillbeof 60marks and internal assessment 15.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

- Thereshallbe**7**questionsinall.

Thefirst questionis **compulsory**comprising 8 short answertypequestions spreadover the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest of the papershall contain three units. Two questions shall be asked from each unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattemptone questionfromeachunit.Eachquestionwillcarry15marks. (15x3=45).

UNIT-I

Collectionofprimarydatathroughquestionnaire(online(google form)/ faceto face)

Errors in surveys: Misunderstanding of question and answers, problem of non-response.

Processing of Data: Questionnaire checking, Editing, Coding, classification, Tabulation, Graphicalrepresentation (Bar charts, Pie Charts, line graphs) Data cleaning, Data Adjusting.

UNIT-II

Analysis of Data and Interpretation: Measures of central tendency (Arithmetic mean median and mode),Dispersion,simplecorrelation(KarlPearsonandSpearmanRankCorrelation)and simpleregression.

UNIT-III

PreparationofaReport:TypesofReport, ResearchReportFormat,PrinciplesofWritingReports, Documentation, Footnotes and References & Bibliography.

Useofcomputersin research.

Ethicsandscientificintegrity:standardsofconduct,privacyindata,Plagiarism.

****PracticalWork:Project Report(collectiondatathroughquestionnaireon aresearch problemand preparation, analysis and interpretation of data).

EssentialReadings:-

1. Bethlehem,J. (2009).*Appliedsurveymethods:Astatisticalperspective.*Wiley.
2. Cochran,W.(2008).*Samplingtechniques,3rded.*Wiley.

SuggestedReadings:-

1. Cooper,D.,Schindler, P.,Sharma, J.(2012).*Businessresearchmethods*,*12thed.* McGraw-Hill.
2. Flick,U.(2012).*Introducingresearchmethodology:Abeginner'sguidetodoinga research project*. Sage Publications.
3. Groves,R.,Fowler, F.,Couper,M.,Lepkowski,J.,Singer,E.,Tourangeau, R.(2009). *SurveyMethodology*.Wiley.
4. Kothari,C.R.(2009).*ResearchMethodsandTechniques*,WileyEastern.
5. Kumar,R.(2014).*Research methodology:Astepbystepguideforbeginners*,*4th ed.* SagePublications.

PUB-MDSC-2

Multi-Disciplinary Courses (MDSC)

Paper: Basics of Indian Administration

Max.Marks:75
Theory:68Marks
Internal Assessment: 07 Marks
Time: 03 Hours

Courseobjective

The objective of this paper is to acquaint the student with the basics of Indian Administration. The paper would introduce the student with the basic features of Indian Administration along with the functioning of Union and State Executive and functioning of various constitutional bodies like UPSC, SPSC etc.

Courseoutcomes

1. To developunderstatingofthebasicfeaturesofIndianAdministration.
2. TogenerateawarenessaboutthefunctioningofUnionandStateExecutive.
3. Todelineatetherelationshipbetweenpoliticalandpermanentexecutive.
4. To understandtheroleofconstitutionalagencies.

Pedagogyofthecourse

The students will be assessed on the basis of written examination comprising of 68 marks and continuous internal assessment of 07 marks based on class test, academic activities,Attendance, Class assignments and class room participations.

INSTRUCTIONSFORPAPER-SETTERSANDCANDIDATES:

- For Private/UniversitySchoolofOpen Learning(USOL)students,whohavenotbeen assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.

ThePaper-Settermustputanoteinquestionpaperinthisregard.

- Thecandidateshallattempt5questionsinall(onecompulsoryandoneeachfromfourunits). Thefirstcompulsoryquestionshallcompriseof10short-answertypequestions,coveringthe wholesyllabus,tobeansweredin25-30wordseach,outofwhichthecandidatewouldbe requiredtoattemptany7.Eachquestionwillcarry2marks.Restofthepapershallcontain3 units,eachunithavingtwoquestions,outof whichthecandidatewouldberequiredtoattempt one. Each question will carry 18marks.

Coursecontents

Features of Indian Administration President:Election,PowersandRole

Unit-I

PrimeMinister:Appointment,PowersandRole

Unit-II

Governor:Appointment,PowersandRole
Chief Minister: Appointment, Powers and Role
Relationship between Political and Permanent Executive

Unit-III

Union Public Service Commission: Composition and FunctionsState PublicServiceCommission:Compositionand FunctionsComptrollerand Auditor General: Appointment and Functions Election Commission of India: Composition and Functions

SuggestedReadings:

Arora, Ramesh K. and Goyal, Rajni (1997) Indian Public Administration: Institutions and Issues. New Age International Publishers: New Delhi

Avasthi,AandAvasthi,AP(2004)IndianAdministration.LaksmiNarainAggarwal:Agra

Fadia,BLandFadia,Kuldeep(2017)IndianAdministration,(NewEdition).Sahitya Bhawan: Agra

Basu, D. D (2013) Introduction to the Constitution of India (21st Edition). Lexus Nexus: New Delhi

Chakraborty,Bidyut(2016)IndianAdministration.Sage:NewDelhi

Ghuman,BS;Monga,Aniland Johal,RamanjitKaur(Eds.)(2012)Corruptionand Qualityof Governance:ExperiencesofSelect CommonwealthCountries. AalekhPublishers:Jaipur

KangleRP(1972)TheKautilyaArthshastra.MotilalBanarsidass:NewDelhi

Kapur,Devesh;Mehta,PratapBhanuandVaishnav,Milan(Eds.)(2017)RethinkingPublic Institutions in India. Oxford University Press: New Delhi

Maheshwari, SR (2000) Indian Administration. Orient Longman: New Delhi Palmer, N D (1961) Indian Political System.

Sharma, Ashok(2016) Administrative Institutions in India. RBSA Publishers: JaipurSharma,

M(2007) Indian Administration.Anmol:NewDelhi

Sharma,PrabhuDattaandSharma,BM(2009)IndianAdministration:Retrospectand Prospect. Rawat Publications: Jaipur

ECOVAC-3: INTRODUCTION TO ENVIRONMENTAL ECONOMICS

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100-199**
CourseDuration/LTP (Lectures, Tutorials,Practical)	**L=30hours(3hoursaweekfor15weeks=01 semester), T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment 10 marks**
Time	**2hours**

CourseObjective:-Thispaperaimstomakeawarestudentsabouttheimportanceofenvironment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

CourseOutcomes:-Thispapercoverstopicsrelatedtoecosystems,naturalresources,biodiversity, conservation environmental pollution and policies.

PedagogyoftheCourseWork:-80%Classroomteaching.

20%InternalAssessment.

.

Instructions for Paper-setter and candidates:

• The maximummarksforthepaperwillbe 50. Thequestionpaperwillbeof 45 marks andcontinuous evaluation 05 marks. Time allowed will be 1.5 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

• Thereshallbe5questionsinall.

The first questioncompulsorycomprising 5 short answer type questions spread overthewholesyllabus.Thecandidatesarerequiredtoattempt 3questions.Each question will carry 5 marks (5x3= 15).

Rest ofthe paper shall contain two units. Two questions shall be asked fromeach unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattempt one question fromeachunit. Each question will carry 15 marks (15x2=30).

UNIT-I

Introduction:Environmentaleconomics:Nature,Scope,andSignificance.Relationbetween Environment and Economy (inter linkages); Environment Kuznets Curve. Commonpropertyresourcesandtheeconomicproblem;EnvironmentalExternalities inProduction and consumption. Managing externalities with taxes and subsidies and Property Rights; Coase Theorem.

UNIT-II

GlobalEnvironmentalPollution:Airandwaterpollution,Acidrain,OzonedepletionandGlobal warming (causes, consequences, and solutions).

Indian Law pertaining to Environmental protection: Environment Protection Act 1986; Air

(PreventionandControlofPollution)Act;Water(PreventionandcontrolofPollution) Act;Wildlife Protection Act; Forest Conservation Act.
ConsumerismandWastemanagement.

EssentialReadings:-

1. Kolstad.C.D.(2011).IntermediateEnvironmentalEconomics.OxfordUniversityPress.
2. Goodwin,N.,Harris,J.,Nelson,J.,Rajkarnikar,P.,Roach,B.,&Torras,M.(2019).Principlesof economics in context. Routledge, Taylor & Francis Group, New York & London.
3. Bharucha,E.(2021).TextbookforEnvironmentalStudies(Thirdedition).OrientBlackswanPvtLtd.
4. Eugine,T.(2005).EnvironmentalEconomics.VarindaPublications,Delhi.

FurtherReadings:-

1. Hussen,Ahmed(2004).PrinciplesofEnvironmentalEconomic(secondedition),Routledge,Taylor & Francis Group, London & New York.
2. Goodstein,E.S.(2002).EconomicsandtheEnvironment(thirdedition)JohnWiley&Sons., Inc.
3. Maddala,G.S.(2004).Microeconomics;Theoryand Practice.McGraw-HillEducation.

ECOVAC-4:INTRODUCTION TO DATA ANALYSIS USING EXCEL-II

Topologyofthecode	**ValueAddedCourse**
LeveloftheCourse	**100-199**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=30hours(2 hours aweekfor15weeks=01semester), T=0, P=0**
Credits	**2**
Max.Marks	**50Marks=Theory40marks+InternalAssessment10 marks**
Time	**2hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy ofthe Course Work: Thecourserelies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporarythemes and issues fromthesyllabi. Specialtutorials/contact hour for one-to-onestudent-teacherinteractions.

Instructions for Paper-setter and candidates:

• The maximummarksforthepaperwillbe 50. Thequestionpaperwillbeof 45 marks andcontinuous evaluation 05 marks. Time allowed will be 1.5 hours.

Thepaper-settermustputanoteinthequestionpaperin thisregard.

• Thereshallbe5questionsinall.

The first questioncompulsorycomprising 5 short answer type questions spread overthewholesyllabus.Thecandidatesarerequiredtoattempt 3questions.Each question will carry 5 marks (5x3= 15).

Rest ofthe paper shall contain two units. Two questions shall be asked fromeach unit andthecandidatesshallbegiveninternalchoice.Thecandidatesshallattempt one question fromeachunit. Each question will carry 15 marks (15x2=30).

Unit-I

Macro RecordingsandMathematicalanalysis
Introduction to Excel Macros. Using Excel-Macro recording for data analyses (without using VB),Solution of matrices.

Unit-II

StatisticalAnalysis
CentralTendencyandDispersionmeasures,Regressionanalysis, Correlationanalysis.

References:

Lloyd,Chris, J. (2011).DataDrivenBusinessDecisions, JohnWiley&Sons.
Levine, D.M., Stephen, D. and Szabat, K.A. (2016). Statistics for Managers Using Microsoft Excel, Pearson Education.
Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall.
Piascik,Chester(2001).AppliedMathematicswithMicrosoftExcel,PennsylvaniaStateUniversity.
Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing.

Suggestions:

1. The ‘spreadsheet operations’ should be mentioned specifically. Some books refer this topic as ‘arithmeticoperation’, while others explain it by referring to commands such as ‘adding column, deleting row and so on.
2. Insemester II,there is imbalancerelatedtodistributionoftopics betweentwo units. Inunit I,thereis a single topic “Macros”, whereas the second unit consists of 5 topics. The outcome would be betterin terms of setting a paper, otherwise there is a chance of higher repetition under question paper.
3. Inotherrespects,thesyllabusinfine,andcovering almosttopicrequiredatthestageofbachelor.

MAT-DSC 151:Algebra and Ordinary Differential Equations

Credits:6(L=5;T=1;P=0)

TotalMarks:150(IncludingInternalAssessment=30) **TimeAllowedforExamination:**3hrs.

InstructionsfortheCandidatesandPaperSetters

Candidateswillbeaskedtoattemptfivequestionsoutofnine,carryingequalmarks.QuestionNo.1spreadover thewholesyllabuswillbecompulsory.

Therewillbetwoquestionsfromeachunitandthestudentswillhavetoattemptonefromeachunit.

Objective: The objectives are tointroduceasystemoflinearequationsandtheirsolutions, polynomialsand their roots and matrix theory. This course will acquaint students to algebra prior totheintroductionofmoreabstract concepts like groups, rings and fields and to exhibit thetechniques for obtaining solutions to ordinarydifferential equations, the basic ideas and the theory behind those techniques. The emphasis will be to combine theabstract concepts with examples in order to intensify the understanding of the subject.

Unit-I

Review of system of linear equations, general theory of system of linear equations, rank of a matrix, row andcolumn ranks of a matrix, system of homogenous and non-homogeneous linear equations, Matrices, Rank of amatrix, Inverse of a Matrix, Elementary linear transformations, Determinants and their properties, Cramer's rule,Cayley-Hamilton Theorem.

Unit-II

A deeper look at Complex Numbers, taking roots of complex numbers, quick review of operations on polynomials,divisors and greatest common divisor, roots of polynomials, fundamentaltheoremofAlgebra(withoutproof),corollaries of fundamental theorem,Evaluating roots of polynomials of third and fourth degree, bounds of roots, Sturm's theorem, Descarte's rule of signs, approximation of roots.

Unit-III

Origin of Differential Equations, Basic definitions, Family of solutions, Geometric interpretation,Isoclines,Initialandboundaryvalueproblem. BasicExistenceTheorem(Statement). Equations of Order One, Separation of variables, Exact Equations, Linear Equations, Integrating factors, Bernoulli's equation,

Unit-IV

General linear equations: General solutions, Linear independenceofsolutions,differential operators,Linearequationswithconstant coefficients:Auxiliaryequation,Non-homogeneous equations: Method of undeterminedcoefficients,Variation of parametermethod,Nonlinear Equations.

(*ScopeasinChapters2,3,4,5,and9of(A),Chapter1,sections2.1-2.6,5.1-5.5,6.1-6.9,7.1-7.3,7.5,8.3,8.5, 9.1-9.4andchapter16of(B)*))

EssentialTextbook

(A) A.KUROSH*HigherAlgebra,*MIRMoscow,1982.

(B) E.D.RAINVILLE,P.E.BEDIENTANDR.E.BEDIENT,*ElementaryDifferentialEquations,* EighthEdition,PearsonIndia,2016.

FurtherReadings

1. SHANTINARAYAN ANDP.K. MITTAL,*AtextbookofMatrices*S.ChandandCompany Limited 2019.

2. S.L.ROSS,*DifferentialEquations,*3rd ed.,JohnWileyandSons,India2004.

3. W. E. BOYCEAND R. C. DIPRIMA, *Elementary differential equationsand boundary value problems*,SeventhEdition,JohnWileyandSons,Inc.,2001

4. E.A. CODDINGTON,*Anintroduction toordinarydifferentialequations,*Prentice- Hall

AECC2: ABILITY ENHANCEMENT COURSE

Course Title:	**AEC–II:LanguageSkills:ReadingandWriting**
Course Code:	ENG (AEC)—031
TypologyofCourse:	AEC
CourseDuration:	60hoursofpracticalinonesemester(4hr/week)
Credits:	2credits
MaximumMarks:	50marks(30Theory+15Practical+5InternalAssessment)

LearningObjectivesoftheCourse:

- Learntocomposecomprehensive andwell-structured paragraphs
- Learntoutilizeliterarytermsandvocabulary,criticalmethods,andvariouslensesof interpretation in their writing
- Toapplyrulesofgrammar
- Tolearntheformattinganddocumentingconventionsofthediscipline

PedagogyoftheCourse:

a) The paper requires 4 hoursofpracticalper week during which students will be taught andexplainedtheaspectsofreadingandwritingthroughpracticaland hands-ontraining.

b) Forthepracticalexam, student willbeaskedtoreadatext, paraphraseagiventext, answer questions and interpret graphs and charts.

c) Thepracticalevaluation,whichcarries15 marks,willbedone byanexternalexaminer either from the same institution/department or from outside.

CourseContents:

UnitI:Reading(Text:*Interact:ACourseinCommunicativeEnglish*,Cambridge)

1. CloseReading
2. Comprehension
3. Summary
4. Paraphrasing
5. InterpretingGraphsandCharts

UnitII.Writing(Text:*WritingSkills*,Cambridge, 03-57)

1. ParagraphWriting
(Available inPrint ande-book:*EffectiveAcademicWritingI&II,* AliceSavage& Patricia Mayer, OUP (USA, 2006)
2. Note-Making
3. Letter/E-mailWriting
4. ResumeWriting
5. ReportWriting

SuggestedReadings:

- *OnTrack:EnglishSkillsforSuccess*, OrientBlackSwan.
- *Embark:EnglishforUndergraduates*,SteveHart,ArvindR.NairandVeenaBharti, Cambridge University Press.
- *Connect:CourseinCommunicativeEnglish*,DebashisBandyopadhyayandMalathi Krishnan, Cambridge University Press.
- *EnrichYourEnglish*,S.R.InthiraandV. Saraswathi,OxfordUniversityPress.
- *CommunicationandLanguageSkills*,SanghitaSen,et al,CambridgeUniversity Press.
- *PracticalEnglish*,K.K.Singh,APHPublishingCorporation.
- *Language,LiteratureandCreativity*,OrientBlackSwan.
- *TheElementsof Style,*WilliamStrunkJr.andE.B.White,Pearson.
- *OnTheArtofWriting*,ArthurQuiller-Couch,NotionPress.
- *FluencyinEnglishPartII*, OxfordUniversityPress.

Testing Scheme

Max.Marks: 50marksTheory:30marks

Practical:15marksInternalAssessment:05marks

T**ime: 3 hours**

Section-A

Q.1.Short Notes.**Twooutoffive**tobeattempted innot morethan150words each. (10marks)

Section-B

Attempt two questions out of given four: (10marks)Q2.Reading Comprehension

Summariseagivenpassage.

Write aparaphraseofthegivenpassage.

InterpretingGraphsandCharts

Section-C

Attemptanytwoquestionsoutof five: (10marks)

ParagraphWriting

Note-Making

Letter/E-mailWriting

ResumeWriting

Report Writing

Syllabi of B.A.(Hons.) Semester-III (under NEP)

ECODSC 5: INTERMEDIATE MICRO ECONOMICS–I

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
CourseDuration/LTP (Lectures,Tutorials,Practical)	**L=60hours(4hoursaweekfor15weeks=01 semester),T=0,P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

CourseDescription

The course is designed to provide a sound training in microeconomic theoryto formally analyze the behaviour of individual agents. Since students are already familiar with the quantitative techniques in the previous semesters, mathematical tools are used to facilitate understanding of the basic concepts. This course looks atthe behaviourofthe consumer and the producerand also covers the behaviour of a competitive firm.

InstructionsforPaper-setterandcandidates:

- The maximum marks for the paper will be 100. The questionpaper will be of80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questions inall.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

Budgetconstraint;propertiesofbudgetset,howbudgetlinechanges. Preference;
Indifference curves, Marginal rate of substitution.
Utility;perfectsubstitutes,perfectcomplements,
Choice;optimalchoice;neutralsandbads,discretegoods, concave preferences.
Demand;incomeoffercurvesandEngelcurves,priceoffercurveanddemandcurve.Revealed preference

UNIT- II

Slutskyequation;the substitutioneffect,thetotalchangeindemand,ratesofchange.
BuyingandSelling;netgrossdemand, offercurvesanddemandcurvesIntertemporalchoice.

UNIT- III

Uncertaintyandrisk aversion
Technology :ProductionFunctions; Homogenousandhomothetic;Cobb DouglasandCES: Properties ;elasticity of substitution ; Eulers theorem.

UNIT-IV

Perfect competition: Assumptions: theory of a firm under perfect competitiondemand and revenue; equilibrium of the firm in the short run and long run; long run industry supply curve: Increasing, decreasing and constant cost industries.
Price controls: Price Ceiling; price floors; production import quotas Welfare: allocative efficiency under perfect competition.

Readings:

1. ScopeasinVarian,H.R(2010).*IntermediateMicroeconomics,a ModernApproach*(8thed.).W.W.NortonandCompany/AffiliatedEast-West Press (India) given as follows;
   Unit1 chapters2to 7
   Unit2chapters 8,9,10
   Unit3 chapters12to 18
   Unit4 chapters19,22,23

AdvancedReadings:

2. Varian.H.R (2010). Intermediate Microeconomics, a Modern Approach (8th ed.)W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.
3. C. Snyder and W. Nicholson.(2010).Fundamentals ofMicroeconomics, Cengage Learning(India).
4. Maddala,G.S.andMiller,R.L.(2006).Intermediate microeconomics:theory,issues,applications.McGraw-HillCompanies.

5. Koutsoyiannis, A.(2003).Modernmicroeconomics.(2nded.).Paperback
6. B. Douglas Bernheim and Michael D. Whinston. (2009). Microeconomics, Tata McGraw-Hill (India).
7. Bergstrom, T.C., &Varian, H.R.(1990).Workouts inintermediate microeconomics. WWNorton.
8. Pindyck, Robert S., and Daniel L. Rubinfeld. Microeconomics. Upper Saddle River, N.J.:Pearson/Prentice Hall, 2009.
9. Mankiw, G. (2000). Microeconomics.In *Principles of Microeconomics* (Vol. 1,pp. 3–6,270–272).

ECODSC6: INTRODUCTORY STATISTICAL METHODS FOR ECONOMIC ANALYSIS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
CourseDuration/LTP (Lectures,Tutorials,Practical)	**L=60hours(4hoursaweekfor15weeks=01 semester),T=0,P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective: The Objective of this paper is to make students familiar with the use of statistical tools in analyzing economic problems.The course content contains simple tools and techniques, considerednecessary for data collection, presentation and analysis with basic idea of probability concepts which are essential for understanding advance topics related to statistical inference.

Course Outcome: On completion of this course students will be able to: (i) Get familiar with various statistical methods used in summarization and analysis of data. (ii) Understand and able to calculate the measures of central tendency and variability. (iii) Understand about the types of correlations and linear regression for bivariate data and their uses in real life problems. (iv) Understand the basic cocepts of probability to be used in statistical inference.

PedagogyoftheCourseWork:80%Classroomteaching.
20%InternalAssessment

InstructionsforPaper-setterandcandidates:

- Themaximummarks forthepaperwillbe100. Thequestionpaperwillbeof80marksand continuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questionsinall.

Thefirstquestion**compulsory**comprising15shortanswertypequestionsspreadoverthewhole syllabus. The candidates are required to attempt 10 questions. Each question shall be of**two** marks (10x2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT–I

Statistics,DataandStatisticalThinking

ScienceofStatistics,Typesofstatisticalapplications,FundamentalelementsofStatistics,Typesof Data, Collecting data. Classification and Tabulation of data, Graphic and Diagrammatic representationof data.The role of statisticsin critical thinking and ethics.

UNIT–II

NumericalDescriptiveMeasures

Measures of Central Tendency: Mean, Median, Mode, GeometricMean andHarmonicMean; *MeasuresofVariability:*Range,QuartileDeviation,Mean DeviationandStandardDeviation: Interpreting the Standard Deviation.

UNIT–III

Techniquestoquantifylinearrelationship

*Correlation and Analysis:*Simple Correlation (Karl Pearson's and Rank Correlation); Partial and Multiple Correlation Analyses;

Regression Analysis: EstimationofaRegressionlineinaBivariateDistribution, Least Squares Method: Interpretation of Correlation Coefficients with help of Regression Coefficients, Coefficient of Determination

UNIT–IV

BasicsofProbability

Sample spaces and events, Definitions of probability (Classical, Empirical and Axiomatic), Unions and intersections, complementary events.

Theadditiveruleandmutuallyexclusiveevents,conditionalprobability,themultiplicative rule and independent events, random sampling and counting rules.

EssentialReadings:

Devore, J. (2012). Probabilityand statistics for engineers, 8th ed. Cengage Learning. (Chapters 1, 2and 12-12.1, 12.2 and 12.5-508 to 511 covering unit III)

JamesMcClave,TerrySincich.(2013).StatisticsforBusinessandEconomics,PearsonPublication. (Chapter 1-1.1to 1.6, Chapter 2-2.1to 2.6, 2.9and 2.10and chapter 3-3.1to 3.8 covering units I, II and IV) **Suggested Readings**

Larsen,R.,Marx,M.(2018),AnIntroductiontoMathematicalStatisticsand itsApplications,Pearson.

Gupta,SC.(8th edition).FundamentalsofStatistics,HimalayaPublishingHouse

.Miller,Irwinand Miller,Marylees(2006):JohnE.Freund'sMathematicalStatisticswithApplications,(7th Edn.),PearsonEducation,Asia.

ECODSC 7:HISTORY OF ECONOMIC THOUGHT

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=60hours(4 hours aweekfor15weeks=01 semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Objective:

Evolution of economic ideas is both a response to contemporary economic problems and a self-conscious attempt to refine earlier ideas to integrate them as a part of current social thought.

Theobjectiveofthiscourse isto familiarizethestudentswiththe historicalevolutionofeconomic ideas into the contemporary economics.

The time-testedmethod ofimparting verbalinstructionsthought lectureswould be used. Examples,so far as possible, would be selected from everyday life-experience. The students would be evaluated at the end of semester through subjective type questions/answers (both short and essay type).

PedagogyoftheCourseWork: 80%Classroomteaching.

20%InternalAssessmentComprisingUnitTestandAssignment/termpaper.

InstructionsforPaper-setterandcandidates:

• Themaximummarksforthepaperwillbe100.Thequestionpaperwillbeof 80marksandcontinuous evaluation 20 marks. Time allowed will be 3 hours.

Thepaper-setter mustputanoteinthequestionpaper inthis regard.

• Thereshallbe9questionsinall.

The first question compulsory comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of two marks (10x2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internalchoice. The candidates shall attempt one question fromeach unit. Each question will carry 15 marks (15x4=60).

UNIT– I

ScopeandSignificanceoftheStudyofHistoryofEconomicThought.

Mercantilism :FactorsResponsiblefortheRiseof Mercantilism; TheMercantilistConception of foreign Trade,State andMoney.

Physiocracy:Rise ofPhysiocracy;ConceptofNetProduct;NaturalOrder;CirculationofWealth.

UNIT– II

ClassicalSchool:AdamSmith:TheoryofValue; Distribution;Growth;Trade. Ricardo: Theory of Value; Distribution; Trade; Views on Machinery.
Malthus:MarketGlutandGlutControversy. J
S Mill: Stationary State.

UNIT– III

KarlMarx:Rise andFallofCapitalism.
NeoClassicalEconomics–Mainfeaturesandcontribution. Keynes: Principle of Effective Demand.
Friedman: Introductionofmonetaryschool.

UNITIV

EvolutionofEconomicthoughtpostworldwar–II.

1960's: The decade of Development;

1970's: Inequality debate(Kuznets etal.)

PhasesofNeo-Liberalism:1980'sand1990's;

Rise of Neo-Nationalism in 2000's.

EssentialReadings:

Jevons,W.S.(1888):TheTheoryofPoliticalEconomy,MacmillanandCo.,NewYorkandLondon. Chapter 1 and 6.
Marshall,(1890):Principles of Economics 8th Edition,Macmillan andCo., NewYorkandLondon Book III, Chapter 6 and Appendix C51.

Pigou,A.C.(1920):TheEconomicofWelfare,MacmillanandCo., NewYorkandLondon,Chapter20.
KeynesJ.M.(1936):TheGeneralTheoryofEmployment,InterestandMoney,ReprintedbyAtlantic Publication, New Delhi. Chapter 2, 3, 8,9,10 and12.
Kalecki M. (1971): Selected Essays on the Dynamics of Capitalism, University Press, Cambridge.
Friedman,M.(1962):CapitalismandFreedom,TheUniversityofChicago Press,Chicago andLondon. Foster, J. B. and: The Endless Crisis, Monthly Review Press. McChesney (2012)
Harvey, D.(2005): ABriefHistoryofNeoliberalism,Oxford UniversityPress,New York.

StegerM.B. and:Neoliberalism: AveryShort Introduction, Oxford Roy, R.K. (2010)UniversityPress, New York.

FurtherReadings:

Screpanti(2006): OutlineofHistoryofEconomicThought.Oxford UniversityPress, Calcutta. Blaug, M (1983): Economic Theory in Retrospect. Vikas Publishing House, New Delhi.
Dasgupta,AK(1986):EpochsofEconomicTheory.OxfordUniversityPress,NewDelhi.
Collander, D C (1984): Neo-Classical Political Economy, Ballinger, Cambridge, M A.
Mandel, E (1971): Marxist Economic Theory. Upra & Co.
Schumpter,JA(1954):HistoryofEconomicAnalysis,OxfordUniversityPress,Oxford.

MAT-DSC-251:Real Analysis

Credits:4(L=5;T=1;P=0)

TotalMarks:100(IncludingInternalAssessment=20) **TimeAllowedforExamination:**3hrs.

InstructionsfortheCandidatesandPaperSetters

Candidateswillbeaskedtoattemptfivequestionsoutofnine,carryingequalmarks.QuestionNo.1spread overthewholesyllabuswillbecompulsory.

Therewillbetwoquestionsfromeachunitandthestudentswillhavetoattemptonefromeachunit.

Objective: To deliver the theoryand applicationsofRiemannintegration,improperintegrals, differentiation, and uniform convergence.

Unit-I

TheRiemannintegral,Riemannintegrablefunctions,fundamentaltheoremofcalculus,the Darboux integral, equivalence theorem.Mean value theorems for the integral calculus.

(Scope:Sections7.1–7.4of(A)).

Unit-II

IntroductiontotheRiemann-Stieltje'sintegral.Improperintegrals,testsforconvergenceof improper integrals, gamma and beta functions.

(Scope:Relevantsectionsfrom (A)andChapter11of(1)).

Unit-III

The derivative of a real valued function, the chain rule, the mean value theorem and its applications, the inter- mediate value property of derivative, L'Hospital's Rules, derivatives of higher order, the Taylor's theorem and its applications.

(Scope:Chapter5of(B)).

Unit-IV

Sequences and series of functions, Discussion of the main problem, Weierstrass' M-test,uniform convergence, uniform convergence and continuity, uniform convergence and limits, uniform convergence and integration, uniform convergence and differentiation, continuous but nowheredifferentiablefunctions,Weierstrassapproximationtheorem(without proof).Power series, interval/radius of convergence.

(Scope:7.1to7.12,7.18,7.26,7.27,3.33,3.38-3.40of(B)).

EssentialTextbooks

(A) R.G.BARTLEANDD.R.SHERBERT,*IntroductiontoRealAnalysis(*3*rded.),*Wiley,2002.

(B) W.RUDIN,*PrinciplesofMathematicalAnalysis(*3*rded.),*McGrawHill,1976.

FurtherReadings

1. S.C.MALIKANDS.ARORA,*MathematicalAnalysis(*3rd*ed),*NewAgeInternational,2008.
2. S.ABBOTT,*UnderstandingAnalysis,*Springer-Verlag,2008.
3. T. M. APOSTOL, *MathematicalAnalysis(*2nd*ed.Reprint),*Narosa,2002.
4. S.K. BERBERIAN, *AFirstCourseinRealAnalysis,*Springer-Verlag,1994.
5. C.G.DENLINGER,*ElementsofRealAnalysis,*Jones&Bartlett(StudentEdition),2011.
6. S. R. GHORPADE AND B.V. LIMAYE, *ACourseinCalculusandRealAnalysis,*Springer,2006.
7. M. H. PROTTERANDC. B. MORREY, *AFirstCourseinRealAnalysis(*2nd*ed.),*Springer,2004.
8. C.C.PUGH,
9. K.A.ROSS,
10. S.C.MALIKANDS.ARORA, *Mathematical Analysis (*3rd *ed.),* New Age International Publishers, 2008.
11. *RealMathematicalA nalysis,*Springer-Verlag,2001.

*ElementaryAnalysis,TheTheoryofCalc ulus,*Springer(SIE),2004.

MDSC 3:SOCIALSTRATIFICATION

Topologyofthecode	**Multidisciplinarycourse**
LeveloftheCourse	**100-199**
CourseDuration/LTP(Lectures, Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01semester), T=0, P=0**
Credits	**3**
Max.Marks	**75Marks=Theory60 marks+InternalAssessment15 marks**
Time	**3hours**

Course Objective: -The systematic study of human behavior and human society is a relatively recent development. Thecentralemphasisofthepaper would beto develop aconceptualclarityregarding the basic sociological terms and themes, thus leading to a development sociological outwork. The course would also discuss the works of foundingfather which led to the developmentof sociological thinking.

Course Outcomes : -1. Students are able to study human behavior and human society relative to recent development. 2. Students develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development of sociological terms and themes, Thusleadingtoadevelopmentofasociologicaloutwork.

PedagogyoftheCourseWork:-80%Classroomteaching.
20%InternalAssessment.

InstructionsforPaper-setter andcandidates:

- Themaximummarksforthepaperwillbe75. Thequestionpaperwillbeof60marksand internal assessment 15.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**7**questionsinall.

The first question is**compulsory**comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest ofthepaper shall containthree units. Two questions shall be asked from eachunit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT –I

SocialStratification:Concept-SocialStratification,InequalitiesandDifferentiation.Elements: Position, Role and Status.

UNITII

Forms of Stratification and Social Mobility: Forms of Stratification – Caste, Class, Race and Gender.SocialMobility:Definition;Indicators–Social,Economic,Political;Types–

Horizontaland Vertical(Upward andDownward)and Intergenerational.

UNIT–III

Social Change: Social Change – Definition. Factors: Demographic, Technological, Educational, Legislative. Processes ofChange: Sanskritization, Westernization, Modernization and Globalization.

EssentialReadings:-

1. TheNewMiddleClassinIndia–ASociologicalAnalysis,Jaipur, Rawat.
2. Singh,Yogendra(1986).*ModernizationofIndianTradition*,Jaipur,Rawat(orlatested.)
3. Srinivas,MN. (2010).*SocialChangeinModernIndia*, OrientBlackSwanPvt.Ltd.
4. Sharma,KL. (2006).*SocialStratificationandMobility*, RawatPublication, NewDelhi.
5. AbrahamFrancis,M.(2004).*ModernSociologicalTheory–AnIntroduction*,Oxford University Press, Delhi.

Suggested Readings:-

1. D'Souza,V.S.(1981).*InequalityandItsPerpetuation*,Manohar Publications,New Delhi.
2. Singh,KS.(ed.)(1992).*EthnicityCasteandPeople:IndiaandSovietUnion.* Manohar Publications, New Delhi.
3. SingerMilton,CohaBeemard(2001). *StructureandChangeinIndianSociety*,Rawat Publications, New Delhi.
4. AtalYogesh.(2006). *ChangingIndianSociety*, RawatPublications,NewDelhi.
5. Beteille,Andre(2009).Sociology–EssaysonApproach&Method.Oxford UniversityPress.
6. *Rao Shankar* C.N. (2012).*Sociology*: principles with an Introduction toSocial Thought.S.Chand and Company Pvt. Ltd.
7. *RaoShankar*C.N.(2012).SOCIOLOGYOFINDIANSOCIETY.S.ChandandCompany Pvt.Ltd.
8. Haralambos,Michael.(2015).Sociology:ThemesandPerspectives.Oxford.
9. Ministryof Women and Child Development, GOI. Report ofthe High Level Committee onStatusofWomeninIndia(2015).http://www.wcd.nic.in/documents/hlc-status-women
10. Johnson,DoleyPaul.(1986).SociologicalTheory(Orlatered.).NewYork,McMillan.
11. Ritzer, George.(2011).SociologicalTheory.London(Orlatested.).McGraw-Hill.
1. CoserLewis.(2004).MastersofSociologicalThought,RawatPublications,NewDelhi.
2. RaoShankar C.N. (2012).Sociology: principles with an Introduction toSocial Thought.S.Chand and Company Pvt.Ltd

Compulsory Punjabi
ਲਾਜ਼ਮੀ ਪੰਜਾਬੀ
PUN-128

Semester- III **ਕ੍ਰੈਡਿਟ-2**

Credit Distribution, Eligibility and Pre-Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course			Eligibility criteria	Pre- requisite of the course (if any)
		Lecture	tutorial	practical		
PUN-128	2	3	1	0		

Credits: 2 L/T/P=3/1/0

Learning objective

- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਨਾਟਕਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਪ੍ਰਾਪਤਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਨਾਟਕਕਾਰਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦੇਨਾਟਕਰਾਹੀਂਨਾਟ-ਕਲਾਤੇਰੰਗਮੰਚਬਾਰੇਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾਦੇਅੰਤਰਸੰਬੰਧਾਂਪ੍ਰਤੀਮੁੱਢਲੀਜਾਣਕਾਰੀਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਦੀਵਿਲੱਖਣਪਛਾਣਤੇਸ਼ਬਦਾਵਲੀਸੰਬੰਧੀਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।

Learning outcomes

- ਇਸਨਾਲਵਿਦਿਆਰਥੀਆਂਅੰਦਰਪੰਜਾਬੀਨਾਟਕਦੀਵਿਧਾਸੰਬੰਧੀਮੁੱਢਲੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਨਾਟਕਦੇਹਵਾਲੇਨਾਲਪੰਜਾਬੀਨਾਟ-ਕਲਾਤੇਰੰਗਮੰਚਦੀਬਣਤਰਸੰਬੰਧੀਵਿਵੇਕਹਾਸਿਲਕਰਪਾਉਣਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਆਂਵਿਚਭਾਸ਼ਾਤੇਉਪ-ਭਾਸ਼ਾਦੇਨਿਖੇੜਸੰਬੰਧੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਆਂਵਿਚਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਦੀਵਿਲੱਖਣਪਛਾਣਸੰਬੰਧੀਸਮਝਪੈਦਾਹੋਵੇਗੀ।

ਕ੍ਰੈਡਿਟ-2
ਨਿਰਧਾਰਤ ਕੋਰਸ ਸਮਾਂ:60 ਘੰਟੇ
ਕੁਲ ਅੰਕ:50
ਲਿਖਤੀ ਅੰਕ:40
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ:10

(ਨੋਟ: ਇਸਕੋਰਸਵਿਚ 2 Unitਹੋਣਗੇ। Unit-1 ਵਿਚਨਾਟਕਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ। Unit-2 ਵਿਚਗੁਰਮੁਖੀਲਿਪੀਦੇਮਹੱਤਵਤੇਪੰਜਾਬੀਭਾਸ਼ਾਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇਅਤੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।)

Unit-I

ਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦਾਨਾਟਕ 'ਵਰ-ਘਰ', ਲਾਹੌਰਬੁੱਕਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930

(ੳ) ਨਾਟਕਦਾਵਿਸ਼ਾ-ਵਸਤੂਤੇਕਲਾਪੱਖ (ਦੋ ਵਿਚੋਂਇੱਕ)
(ਅ) ਕਿਸੇਪਾਤਰਦਾਪਾਤਰ-ਚਿਤਰਣ (ਦੋ ਵਿਚੋਂਇਕ)
(ੲ) ਨਾਟਕਵਿਚੋਂਲਘੂਪ੍ਰਸ਼ਨ (ਦਸ ਵਿਚੋਂਪੰਜ)

(5+5+10=20)

Unit-II

ਗੁਰਮੁਖੀਲਿਪੀਅਤੇਪੰਜਾਬੀਭਾਸ਼ਾਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂ

(ੳ) ਗੁਰਮੁਖੀਲਿੱਪੀਦਾਮਹੱਤਵਅਤੇਪੰਜਾਬੀਭਾਸ਼ਾਲਈਇਸਦੀਸਾਰਥਕਤਾ (ਦੋ ਵਿਚੋਂਇਕ)
(ਅ) ਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾ: ਅੰਤਰਸੰਬੰਧਤੇਨਿਖੇੜਾ (ਦੋ ਵਿਚੋਂਇੱਕ)
(ੲ)ਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂ (ਸੱਤਾਵਿਚੋਂਪੰਜ)

(5+5+10=20)

ਨੋਟ: ਲਾਜ਼ਮੀਪੰਜਾਬੀਵਿਸ਼ੇਦੇਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇਅਤੇਪੇਪਰ 50 ਅੰਕਾਂਦਾਹੋਵੇਗਾ। ਵਿਦਿਆਰਥੀਹਫ਼ਤੇਵਿਚਘੱਟੋ-ਘੱਟਪੰਜਦਿਨਪੜਨਗੇ।

ਪੇਪਰਸੈਟਰ/ਅਧਿਆਪਕ ਲਈ ਹਦਾਇਤਾਂ ਅਤੇ ਪ੍ਰਸ਼ਨ-ਪੱਤਰ ਦੀ ਰੂਪ-ਰੇਖਾ:-

1. ਪ੍ਰਸ਼ਨ-ਪੱਤਰਦੇਕੁੱਲ 50 ਅੰਕਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟਦੇਬਰਾਬਰਹੈਅਤੇਕੋਰਸਦਾਨਿਰਧਾਰਤਸਮਾਂ 60 ਘੰਟੇਹੋਵੇਗਾ।
2. ਲਿਖਤੀਪ੍ਰਸ਼ਨ-ਪੱਤਰ 40 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।
3. ਇੰਟਰਨਲਅਸੈਸਮੈਂਟਦੇ 10 ਅੰਕਹੋਣਗੇ।
4. ਪੇਪਰਦਾਸਮਾਂਤਿੰਨਘੰਟੇਹੋਵੇਗਾ।

ਯੂਨਿਟਅਤੇਅੰਕਾਂਦੀਵੰਡ

1. ਯੂਨਿਟਇਕਵਿਚ 'ਵਰ-ਘਰ' ਨਾਟਕਵਿਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਹਯੂਨਿਟ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।

(ੳ) ਇਸਭਾਗਵਿਚਨਾਟਕਦੇਵਿਸ਼ਾ-ਵਸਤੂਅਤੇਕਲਾਪੱਖਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।
(ਅ) ਇਸਭਾਗਵਿਚਪਾਤਰ-ਚਿਤਰਣਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।
(ੲ) ਇਸਭਾਗਵਿਚਨਾਟਕਸੰਬੰਧੀਲਘੂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚਕੁਲ 14 ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਚਾਰਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ, ਜੋ ਪੰਜ-ਪੰਜਅੰਕਾਂਦੇਹੋਣਗੇ।ਬਾਕੀ ਦਸ ਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿ ਦੋ-ਦੋ ਅੰਕਦੇਹੋਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁੱਲਅੰਕ 20 ਹੋਣਗੇ।

2. ਯੂਨਿਟ ਦੋ ਵਿਚਗੁਰਮੁਖੀਲਿਪੀਦੇਮਹੱਤਵਅਤੇਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।

(ੳ) ਇਸਭਾਗਵਿਚਗੁਰਮੁੱਖੀਲਿਪੀਦੇਮਹੱਤਵਤੇਸਾਰਥਕਤਾਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਅ) ਇਸਭਾਗਵਿਚਭਾਸ਼ਾਅਤੇਉਪ-ਭਾਸ਼ਾਦੇਅੰਤਰ-ਸੰਬੰਧਤੇਨਿਖੇੜੇਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ੲ) ਇਸਭਾਗਵਿਚਪੰਜਾਬੀਦੀਆਂਉਪ-ਭਾਸ਼ਾਵਾਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚਕੁੱਲ 11 ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਚਾਰਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ ਜੋ ਪੰਜ-ਪੰਜਅੰਕਾਂਦੇਹੋਣਗੇ।ਬਾਕੀ ਦਸ ਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿ ਦੋ-ਦੋ ਅੰਕਾਂਦੇਹੋਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

ਨਿਰਧਾਰਤ ਪੁਸਤਕਾਂ

1. ਵਰ-ਘਰ, ਈਸ਼ਵਰਚੰਦਰਨੰਦਾਦਾਨਾਟਕ , ਲਾਹੌਰਬੁੱਕਸ਼ਾਪ, ਲੁਧਿਆਣਾ, 1930
2. ਪੰਜਾਬੀਨਾਟਕਦਾਇਤਿਹਾਸ, ਸਤੀਸ਼ਕੁਮਾਰਵਰਮਾਂ (ਡਾ.), ਪੰਜਾਬੀਸਾਹਿਤਅਕਾਦਮੀ, ਦਿਲੀ, 2005
3. ਪੰਜਾਬੀਨਾਟਕਅਤੇਰੰਗਮੰਚ, ਕਮਲੇਸ਼ਉੱਪਲ, ਪੰਜਾਬੀਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ, 2004
4. ਕਾਲਜਪੰਜਾਬੀਵਿਆਕਰਣ, ਡਾ.ਹਰਕੀਰਤਸਿੰਘਅਤੇਗਿਆਨੀਲਾਲਸਿੰਘ, ਪੰਜਾਬਸਟੇਟਯੂਨੀਵਰਸਿਟੀਟੈਕਸਟਬੁੱਕਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
5. ਪੰਜਾਬੀਭਾਸ਼ਾਦਾਵਿਆਕਰਨ, ਜੋਗਿੰਦਰਸਿੰਘਪੁਆਰਅਤੇਹੋਰ, ਪੰਜਾਬੀਭਾਸ਼ਾਅਕਾਦਮੀ, ਜਲੰਧਰ 1999
6. ਪੰਜਾਬੀਭਾਸ਼ਾਦਾਵਿਆਕਰਨ, ਜੋਗਿੰਦਰਸਿੰਘਪੁਆਰਅਤੇਹੋਰ, ਪੰਜਾਬੀਭਾਸ਼ਾਅਕਾਦਮੀ, ਜਲੰਧਰ 1997
7. ਪੰਜਾਬੀਵਿਆਕਰਨ: ਸਿਧਾਂਤਅਤੇਵਿਹਾਰ, ਬੂਟਾਸਿੰਘਬਰਾੜ (ਡਾ.), ਚੇਤਨਾਪ੍ਰਕਾਸ਼ਨ, ਲੁਧਿਆਣਾ

MAT-SEC-156: Working with Mathematical Softwares

(Mathematica/Matlab/Octave/Maple)

Credits: 3(L=0;T=0;P=3) Totalhours: 90

(Practicals=90)Total Marks: 75 (Including Internal Assessment= 15) TimeAllowedfor Examination: 3hrs.

InstructionsfortheCandidatesandPaperSetters

Candidateswillbeaskedtoattemptfivequestions outofnine,carryingequalmarks.QuestionNo.1 Spreadover thewholesyllabuswillbecompulsory.

Therewillbetwoquestionsfromeachunitandthestudentswillhavetoattempt onefromeachunit.

CourseOutcome:BasicknowledgeofsoftwareMathematica,Matlab,Octaveand/orMaple.

ListofPracticals

Unit-I

1. Matrixoperations(addition,multiplication,inverse,transpose,determinantandrank).
2. Plottingofgraphsofpolynomialsofdegree4and5,thederivativegraph,thethesecondderivativegraphand comparing them.
3. Plotting functions e^{ax+b};log(ax+b); 1=(ax+b);sin(ax+b);cos(ax +b);jax+bjandtoillustratetheeffect ofaandb on the graph.
4. Obtainingsurfacesofrevolutionofcurves.

Unit-II

1. TracingofconicsinCartesiancoordinates/polarcoordinates.3Dplots.
2. Computationoflimits,derivativesandintegrationofvectorfunctions.
3. Tangentplanestosurfacesatagivenpoint.
4. Sketchingellipsoid,hyperboloidofoneandtwosheets,ellipticcone,elliptic,paraboloid,hyperbolic paraboloid using Cartesian coordinates.

Unit-III

1. Plottingthesolutionsoffirstorderdifferentialequation.
2. Plottingthesolutionsofsecondorderdifferentialequation.
3. Plottingthesolutionsofthirdorderdifferentialequation.

Unit-IV

1. Solutionofinitialvalueproblems.
2. Solutionofboundaryvalueproblem.
3. Exponentialgrowthmodel.
4. Exponentialdecaymodel.
5. Limitedgrowthofpopulation.

EssentialTextbooks

(A) GeorgeB.Thomas,MauriceD.WeirandJoelR.Hass,Thomas' Calculus, 12th Ed., PearsonEducation, 2014.

(B) Belinda Barnes and Glenn R. Fulford, Mathematical Modeling withCaseStudies,ADierentialEquationApproachusingMaple and Matlab, Second Edition, Taylor and Francis group, London and 2009.

r(dC)MarthaL.AbellandJames P.Braselton,DierentialEquations withMATHEMATICA,3ed.,ElsevierAcademicPress,2004.

Further Readings

1. JosephL.TaylorFoundationsofAnalysis,PureandApplied Undergraduate Texts, 18, AmericanMathematical Society, Providence, RI, 2012.
2. ShantiNarayan,IntegralCalculus,S.ChandandCompanyLtd,2001.
3. M.J.Strauss,G.L.BradleyandK.J.Smith,Calculus,3rdEd., DorlingKindersley(India)P.Ltd.(Pearson Education),2007.
4. H.Anton,I.BivensandS.Davis,Calculus,7thEd.,JohnWileyandSons, 2002.
5. R. Courant and F. John, Introduction to Calculus and Analysis(VolumesI&II),Springer-Verlag,Inc.,1989.
6. S.L.Ross,DierentialEquations,3rded.,JohnWileyandSons,India2004.
7. E.A.Coddington,Anintroductiontoordinarydierentialequations, Prentice- Hall of India, 1961.
8. C.H.EdwardsandD.E.Penny,DierentialEquationsand Boundary Value problemsComputing and Modeling, PearsonEducationIndia,2005.
9. W. E. Boyce and R. C. Diprima, Elementary di erential equationsandboundaryvalueproblems,SeventhEdition, John Wiley and Sons, Inc., 2001
10. EarlD.RainvilleandP.E.Benediet,Elementarydierential equations, Seventh Edition,Macmillian, Publishing Company, 1989.

Syllabi of B.A.(Hons.) Semester-IV(under NEP)

ECODSC8:INTERMEDIATE MICRO ECONOMICS– II

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=60hours(4hoursaweekfor15 weeks= 01semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+Internal Assessment20 marks**
Time	**3hours**

CourseDescription

This course is a sequel to Intermediate Microeconomics I. The emphasis will be on giving conceptual clarity to the student coupled with the use of mathematical tools and reasoning.It covers general equilibrium and welfare, imperfect markets and topics underinformation economics.

InstructionsforPaper-setterandcandidates:

- The maximum marks for the paper will be 100.The question paperwillbe of 80marks andcontinuousevaluation20marks. Time allowedwillbe 3 hours.

Thepaper-settermustputanoteinthequestionpaperinthisregard.

- Thereshallbe**9**questions inall.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from eachunit andthe candidatesshallbegiveninternalchoice. Thecandidatesshall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT-I

ImperfectCompetition

Monopoly;pricing;measurementofmarketpower;

Multiplant monopoly Price discrimination; Inter temporal; peak-load pricing; two-part tariff

Regulation of Monopoly

Monopolisticcompetition:priceand output determination Economic efficiency and resource allocation

UNIT-II

ImperfectCompetition

Oligopoly;Priceleadership;Cournot,StacklebergandBertrandModels

Gametheoryandcompetitivestrategies:Dominant Strategy;Mixedstrategies;Prisoners Dilemma ;Repeated andSequentialgames

UNIT-III

General Equilibrium, Efficiency and Welfare

Equilibriumand efficiency under pure exchange and production;overallefficiencyWelfareeconomics;Social welfare, welfare maximisation, envy and equity,

UNIT-IV

MarketFailure

Externalities; Productionand Consumption, CoaseTheorem, Market Signals, Public Goods: Provision of Public Goods,Free Riding,DifferentLevels of PublicGoods.Asymmetric Information: Market for Lemons, Moral Hazard, Adverse Selection, Signaling.

Readings:

1.ScopeasinVarian.H.R(2010).*IntermediateMicroeconomics,a ModernApproach*(8thed.)W.W.NortonandCompany/AffiliatedEast-West Press (India) given as follows:

Unit1chapters24,25
Unit2chapters27,28
Unit3 chapters31,32,33
Unit4 chapters34,36,37

AdvancedReadings:

1. Varian.H.R (2010). *Intermediate Microeconomics, a Modern Approach*(8th ed.)W.W. Norton and Company/Affiliated East-West Press (India).The workbook by Varian and Bergstrom could be used for problems.
2. C. Snyder and W. Nicholson.(2010).*Fundamentals of Microeconomics,*Cengage Learning (India).
3. Maddala, G.S. and Miller, R. L. (2006). *Intermediate microeconomics: theory, issues, applications*. McGraw-Hill Companies.
4. Koutsoyiannis,A.(2003).*Modernmicroeconomics.(2nd ed.).*Paperback
5. B. Douglas Bernheim and Michael D. Whinston. (2009). Microeconomics Tata McGraw-Hill(India).
6. Bergstrom,T.C.,& Varian,H.R. (1990).Workouts in intermediate microeconomics.WWNorton.
7. Pindyck,RobertS.,andDanielL.Rubinfeld.Microeconomics.Upper SaddleRiver,N.J.:Pearson/Prentice Hall, 2009.
8. Mankiw,G.(2000). Microeconomics.In*Principlesof Microeconomics* (Vol. 1,pp.3–6,270–272).

ECODSC9-INTERMEDIATE MACROECONOMICS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=60hours(4 hours aweekfor15weeks=01 semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

CourseObjective

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregated analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essentialfor the propercomprehensionofthe different issuesand policies. Macroeconomicsnow is not only a scientific method of analysis; but also a bodyof empiricaleconomic knowledge.

The paper entitled “Intermediate Macroeconomics” equips the students at the undergraduatelevel to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

PedagogyoftheCourseWork

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporarythemes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

After completion of the course, a student must be capable of understanding the determinants of moneydemandandsupply. Also thepapershallinculcatetheabilitiesamong studentsto develop open economy IS-LM based models to work out the key macroeconomic targets. An understanding shall be developed on behaviour of general price level and its use as a key policy instrument to target real variables such as unemployment and output. The theories of explaining trade cycles shall make a student capable of understanding nature and causes of fluctuations in economic activity.

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.

2. Thereshallbe9questionsinall.

3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each).

It would carry15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question fromeach Unit— 4 in all. Each question shall carry15 marks ($15 \times 4 = 60$).

Unit-I

(MoneySupplyand Demand)

***Supply of Money*:** Theoretical Debate and Empirical Attempts to define money; Components of Supply of Money, Credit Creation by Commercial Banks, Money Multiplier.

***Demand for Money*:** Classical Quantity Theory, Keynesian Theory, Baumol and Tobin's Contributions. Friedman's Restatement of Quantity Theory of Money.

Unit-II

(SimultaneousEquilibrium&TatonnementProcessunderClosedandOpenEconomies)

IS and LM Framework: *Derivation, Properties, Shifts and Rotations of IS and LM Curves under closed and open economy systems. Derivation, Properties, Shifts, and Rotations of BP Curve. Simultaneous Equilibrium in Money and Product Markets. Impact of Opening-up on simultaneous equilibrium (i.e., IS-LM-BP simultaneous equilibrium).*

MonetaryandFiscalPolicies: *Objectives,ConflictsamongObjectives.RelativeEffectivenessof Monetary and Fiscal Policies under Different Situations in IS-LM-BP Framework. International Trinity and Quadrilemma choices under IS-LM-BP framework.*

Unit-III

(InflationTheoriesandPhilipsCurve)

Inflation: *Effects of Inflation. Theories of Inflation: Quantity Theory, Keynesian Theory, Monetarist views on Inflation, Modern theory of Inflation, Structural Theory.*

The menu of policy choices: *Philips Curve Analysis –Short Run and Long Run views. The Monetarist-Keynesian Debate and the Phillips Curve.*

UNIT-IV

(TradeCycleModels/Theories)

Trade Cycle: *Hansen–Samuelson Accelerator–Multiplier Interaction Model, Hicks Model, Kaldor Model, Goodwin model of endogenous cycles.*

RecommendedReadings:

Ackley,G.(2007).MacroeconomicTheory***. Macmillan.***
Allen,R.G.D.(1967). Macro-EconomicTheory:AMathematicalTreatment***.PalgraveMacmillan.***
Beckerman,W.(1980).IntroductiontoNationalIncomeAnalysis, *LittlehamptonBookServicesLtd.*
Blanchard,O.(2009).Macroeconomics***.NewDelhi:PearsonEducationInc.***
Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics***, Delhi: PHI Learning Pvt. Ltd. Branson, W. H. (1989).*** Macro Economic Theoryand Policy ***(3rd ed.). Harper Collins.***Cooper,R.,Edey,HaroldC.,andPeacock,A.T.(2003). NationalIncomeandSocialAccounting. *Routledge.*

D'Souza,E.(2008).Macroeconomics. ***PearsonEducation.***

Dornbusch,R.,Fisher,S&Startz, R.(2002).Macroeconomics***, (9thed.). NewDelhi: Tata***

McGrawHills.

Levacic,R.&Rebmann,A.(1982). Macroeconomics:AnIntroductionto Keynesian Neoclassical Controversies***. Macmillan.***

Miller,R.L.&Pulsinelli,R.(1986).Macroeconomics.***NewYork:Harper&Row. Ola Olsson (2010).*** Essentials of Advanced Macroeconomic Theory***, Routledge.***

Romer,D.(1996).AdvancedMacroeconomics***.McGraw-Hill.***

UmaDattaRoyChoudhury(2000).NationalIncomeAccounting,*Macmillan*.

FurtherReadings:

Abel,A.B.,Bernanke,B.S.(2007). Macroeconomics.***NewDelhi:PearsonEducationInc.***

Dornbusch, R. (1980). Open Economy Macroeconomics***. New York: WeltwirtschaftlichesArchive.***

Frisch,H.(1983).TheoriesofInflation***.Cambridge:CambridgeUniversityPress.***

Froyen, R. T. (2002). Macro Economics. ***New Delhi: Pearson Education Inc.***

Gordon,R.J.(2012).Macroeconomics***. Delhi:PHILearningPvt.Ltd.***

Heijdra,B.J.(2017).TheFoundationsofModernMacroeconomics***.OxfordUniversityPress.***

Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle ***Oxford*****:** Clarendon Press***.***

Lucas,R.(1981).StudiesinBusinessCycleTheory***,Massachusetts:MITPressCambridge. Makin, A. J. (2002).*** International Macroeconomics***. Princeton Hall.***

Mankiw,N. G.&Romer,D.(1991).NewKeynesianEconomics.MIT,2Volumes.

Mathews, R. C. O. (1964). The Trade Cycle. ***Cambridge: Cambridge University Press.***

Pentecost,E.(2000). Macroeconomics–AnOpenEconomyApproach. ***MacMillanPressLtd.***

Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. ***New Age International Publishers.***

Sherffrin,S.(1999).RationalExpectations.***CambridgeUniversityPress.***

Snowden,B&Vane,H.R.(2005). ModernMacroeconomics:ItsOrigin,Development and Current Stat., ***UK: Edward Press.***

Wendy,Carlin/Soskice,D.(2007).Macroeconomics:Imperfections,InstitutionsandPolicies***. Oxford University Press.***

ECODSC10–IntroductoryEconometrics

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=60hours(4 hours aweekfor15weeks=01 semester), T=0, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

CourseDescription

Objective: Application of economic theory need a reasonable understanding of economic relationship and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equations models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive useofblackboard. Examples, in so far as possible, would be selected fromeveryday life/experience. The feedback would be sought through the students writing assignments and written examinations.

InstructionsforPaper-setterandcandidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks andcontinuous evaluation 20 marks. Time allowed will be 3 hours. The paper-setter must put a note in the question paper in this regard.
- There shall be **9** questions in all. The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each questionshall beof**two** marks(10 x2=20) Rest of thepaper shallcontain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

Unit-I

Introduction: NatureandScopeofEconometrics,MethodologyofEconometrics.

Specification of an econometric model. Nature of Data used in estimation.

Simple Regression Models: Estimation of model by method of Ordinary Least Squares; PropertiesoftheLeast Squaresestimators, theGauss-Markovtheorem. Goodness offit -R Square & Adjusted R Square. Testing the significance of Regression Coefficients.

UNIT-II

Multiple Regression Models: Estimation of model by method of Ordinary Least Squares; Properties of the Least Squares estimators, the Gauss-Markov theorem.

Specification Errors: Types of Specification errors. Consequences and Remedial measures for Omission of a relevant variable; and Inclusion of irrelevant variable.

UNIT-III

ErrorsofMeasurement:ErrorsofMeasurementinDependentandIndependent variables.

EncompassingandNon-EncompassingModels: Introduction.ModelSelectionCriteria.

Unit-IV

Econometrics Problems: Nature, Consequences, Detection and Remedial measures of the problems of Multicollinearity, Heteroscedasticity and Autocorrelation.

EssentialReading:

GujaratiDamodarN.(2004):BasicEconometrics(4thed.),Singapore.McGrawHillsCo.

FurtherReadings:

Green,William(2000):EconometricAnalysis(4thEd.),PearsonEducation.

Johnston,J&Dinardo,J. (2007):Econometric(4thEd.), McGrawHillsCo.

Maddala,GS(2007):IntroductiontoEconometrics,3rdEdition,JohnWiley&Sons,Ltd., England.

Pindyck, R S & Rubinfeld, D L (1998): Econometric Models & Economic, Forecasts, McGrawHills Co.

Intriligator, M J & Bodkin, H (1996):Econometric Models Techniques and Application. Prentice Hall.

ECODSC11-GROWTH MODELS

Topologyofthecode	**Major/Minor**
LeveloftheCourse	**100-199**
CourseDuration/LTP (Lectures,Tutorials,Practical)	**L=60hours(4hoursaweekfor15weeks=01 semester),T=0,P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective

A continuous economic growth is the primary condition for economic development. The determinantsofgrowthdiffer onthe basis ofthe stageofeconomic development acountryfaces. Thus, a studyofthe macroeconomic GrowthModels is necessaryto understand the determinants of economic growth. The idea and analysis of growth has therefore, assumed such a great significance in recent times that a prior understanding of growth theories is considered essential for the proper comprehension of the different macroeconomic issues and policies. Growthmodels now are not only a scientific method of economic analysis; but also a body of empirical economic knowledge.

The paper entitled "GrowthModels" equipsthe studentsatthe undergraduatelevel to understand systemic facts and latest theoretical developments for theoretical and empirical analysis of growth dynamics.

The students will be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts will be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

PedagogyoftheCourseWork

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assigntopic for assignments oncontemporarythemes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Course Outcomes

The knowledge of theories and growth models would help the students better understand the entire economic system and its growth over time. The basic models mainly focus on the contribution of inputs to production while assuming the technology is constant. The course also covers measuring the level of technology and its role in changing economic structure. The importance of main fundamentals such as public infrastructure, institutions, and efficientresourceallocationhasalso beencoveredto understandcurrent development issues. Overall, this course would lay the foundation for the students to understand the real issues in economic development.

INSTRUCTIONSFORTHEPAPER-SETTERAND CANDIDATES

1. Thesyllabusofthispaper hasbeendividedinto four units.
2. Thereshallbe9questionsinall.
3. The first question, which would be compulsory, shall be short answer type word limit (25-30 each).

It would carry15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question fromeach Unit— 4 in all. Each question shall carry15 marks ($15 \times 4 = 60$).

Unit-I
(InitialDiscussiononGrowth Theory)

Constant capital output ratio based growth models: Domar model of growth, Harrod Model, Similarities between two models, Stability property of Harrod and types of growth, Deviations from knife edge and possible mismatches among three types of growth rates.

Unit-II
(ProductionFunctionandGrowth Theories)

Neo-Classical Model including Solow Model; conditions of steady state, deviation from steady state. Solow residualas technological progress, the Golden rule. **Overlapping Generations and Dynamic Efficiency**

Unit-III
(EconomicGrowthand Convergence)

The Convergence Hypothesis; Beta and *sigma* convergences, idea and empirical evidences.Some advance measures of convergence; Club and Gamma convergences, some empirical evidences.

Unit-IV
(EndogenousGrowth)

Limitations of neoclassical growth theories and need of endogenising growth. The AK model as initial endogenous growth model. Neoclassical endogeneous growth; capital accumulation, externalities and human capital. **Directed Technical Change: Endogenous Skill-Bias and Endogenous Labor-Augmenting Technological Change. Competition and technological change.**

RecommendedReadings:

Acemoglu, Daron(2009).*IntroductiontoModernEconomicGrowth*. PrincetonUniversityPress, 2009. ISBN: 9780691132921.

Freitas, M.L. (2022). *Economic Growth Models: A Primer / Student's Guide*, Nova School of Business and Economics, Online Available: https://sweet.ua.pt/afreitas/growthbook/MLFGrowthNotesnet.pdf

Jones,Hywel G.(1976).*An Introductionto ModernTheories ofEconomicGrowth*,McGraw- Hill, Republished in 1976.

Solow,Robert(1970).GrowthTheory:AnExposition,ClarendonPress,Oxford,UK.

Compulsory Punjabi
ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ
PUN-128

Semester- IV **ਕ੍ਰੈਡਿਟ-2**

Credit Distribution, Eligibility and Pre-Requisites of the Course

Course title & Code	Credits	Credit distribution of the Course Lecture tutorial practical			Eligibility criteria	Pre-requisite of the course (if any)
PUN-128	2	3	1	0		

Credits: 2 L/T/P=3/1/0

Learning objective

- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਦੇਵਿਸ਼ੇਤੇਕਲਾਪੱਖਸੰਬੰਧੀਮੁੱਢਲਾਗਿਆਨਹਾਸਿਲਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਨਿਬੰਧਲੇਖਣਦੇਹੁਨਰਨੂੰਗ੍ਰਹਿਣਕਰਨਗੇ।
- ਇਸਕੋਰਸਰਾਹੀਂਵਿਦਿਆਰਥੀਵਿਹਾਰਕਸੂਚਨਾਹਿਤਨੋਟਿਸਤੇਮੁਹਾਵਰਿਆਂਦੀਵਰਤੋਂਸੰਬੰਧੀਜਾਣਕਾਰੀਪ੍ਰਾਪਤਕਰਨਗੇ।

Learning outcomes

- ਇਸਨਾਲਵਿਦਿਆਰਥੀਅੰਦਰਪੰਜਾਬੀਕਹਾਣੀਲੇਖਣਸੰਬੰਧੀਮੁੱਢਲੀਸੂਝਪੈਦਾਹੋਵੇਗੀ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਪੰਜਾਬੀਕਹਾਣੀਦੇਵਿਸ਼ਿਆਂਨੂੰਸਮਝਣਦੇਯੋਗਬਣਨਗੇ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਨਿਬੰਧਲੇਖਣਦੀਬਣਤਰਨੂੰਸਮਝਣਦੇਯੋਗਬਣਨਗੇ।
- ਇਸਨਾਲਵਿਦਿਆਰਥੀਵਿਹਾਰਕਵਰਤੋਂਵਿਚਆਉਣਵਾਲੇਸੂਚਨਾਹਿੱਤਨੋਟਿਸਤੇਮੁਹਾਵਰਿਆਂਦੀਵਰਤੋਂਸੰਬੰਧੀਸੂਝਪ੍ਰਾਪਤਕਰਪਾਉਣਗੇ।

ਕ੍ਰੈਡਿਟ-2
ਨਿਰਧਾਰਤਕੋਰਸ ਸਮਾਂ:60 ਘੰਟੇ
ਕੁਲ ਅੰਕ:50
ਲਿਖਤੀ ਅੰਕ:40
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ:10

(ਇਸਕੋਰਸਵਿਚ 2 Unit ਹੋਣਗੇ। Unit-1 ਵਿਚਨਿਰਧਾਰਿਤਕਹਾਣੀਸੰਗ੍ਰਹਿ 'ਕਥਾ-ਕਹਾਣੀ' ਵਿੱਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ। ਯੂਨਿਟ ਦੋ ਵਿਚਵਿਹਾਰਕਪੰਜਾਬੀਨਾਲਸੰਬੰਧਿਤਸਵਾਲਪੁੱਛੇਜਾਣਗੇ, ਜੋ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।)

Unit-1

ਪੁਸਤਕ 'ਕਥਾ-ਕਹਾਣੀ' (ਕਹਾਣੀਸੰਗ੍ਰਹਿ), ਡਾ. ਧਨਵੰਤਕੌਰ (ਸੰਪਾ.), ਪਬਲੀਕੇਸ਼ਨਬਿਊਰੋ, ਪੰਜਾਬਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

(ੳ) ਕਹਾਣੀਦਾਵਿਸ਼ਾ-ਵਸਤੂਤੇਮੰਤਵ (ਦੋ ਵਿਚੋਂਇੱਕ)

(ਅ) ਕਹਾਣੀਦੇਕਿਸੇਪਾਤਰਦਾਪਾਤਰ-ਚਿਤ੍ਰਣ(ਦੋ ਵਿਚੋਂਇੱਕ)

(ੲ) ਪੁਸਤਕਵਿਚੋਂਲਘੂਪ੍ਰਸ਼ਨ: ਕਹਾਣੀਆਂਤੇਲੇਖਕਾਂਦੇਸੰਬੰਧਵਿਚ (ਦਸ ਵਿਚੋਂਪੰਜ)

(5+5+10=20 ਅੰਕ)

Unit-2ਵਿਹਾਰਕਪੰਜਾਬੀ

(ੳ) ਨਿਬੰਧਲੇਖਣ: ਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਿਤ (ਦੋ ਵਿਚੋਂਇੱਕ)

(ਅ) ਮੁਹਾਵਰਿਆਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂ(ਸੱਤਵਿਚੋਂਪੰਜ)

(ੲ) ਅਖਾਣਾਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂ (ਸੱਤਵਿਚੋਂਪੰਜ)

(10+5+5 =20 ਅੰਕ)

ਨੋਟ: ਲਾਜ਼ਮੀਪੰਜਾਬੀਵਿਸ਼ੇਦੇਕ੍ਰੈਡਿਟ 2 ਹੋਣਗੇਅਤੇਪੇਪਰ 50 ਅੰਕਾਂਦਾਹੋਵੇਗਾ। ਵਿਦਿਆਰਥੀਹਫ਼ਤੇਵਿਚਘੱਟੋ-ਘੱਟਪੰਜਦਿਨਪੜ੍ਹਨਗੇ।

ਪੇਪਰਸੈਟਰ/ਅਧਿਆਪਕਲਈਹਦਾਇਤਾਂਅਤੇਪ੍ਰਸ਼ਨ-ਪੱਤਰਦੀਰੂਪ-ਰੇਖਾ:-

1. ਪ੍ਰਸ਼ਨ-ਪੱਤਰਦੇਕੁੱਲ 50 ਅੰਕਹਨ ਜੋ ਕਿ 2 ਕ੍ਰੈਡਿਟਦੇਬਰਾਬਰਹੈਅਤੇਕੋਰਸਦਾਨਿਰਧਾਰਤਸਮਾਂ 60 ਘੰਟੇਹੋਵੇਗਾ।
2. ਲਿਖਤੀਪ੍ਰਸ਼ਨ-ਪੱਤਰ 40 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।
3. ਇੰਟਰਨਲਅਸੈਸਮੈਂਟਦੇ 10 ਅੰਕਹੋਣਗੇ।
4. ਪੇਪਰਦਾਸਮਾਂਤਿੰਨਘੰਟੇਹੋਵੇਗਾ।

ਯੂਨਿਟਅਤੇਅੰਕਾਂਦੀਵੰਡ

1. ਯੂਨਿਟਇਕਵਿਚ 'ਕਥਾ-ਕਹਾਣੀ' (ਕਹਾਣੀਸੰਗ੍ਰਹਿ) ਪੁਸਤਕਵਿਚੋਂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ। ਇਹਯੂਨਿਟ 20 ਅੰਕਾਂਦਾਹੋਵੇਗਾ।

(ੳ) ਇਸਭਾਗਵਿਚਕਹਾਣੀਦੇਵਿਸ਼ਾ-ਵਸਤੂਅਤੇਮੰਤਵਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਭ) ਇਸਭਾਗਵਿਚਪਾਤਰ-ਚਿਤਰਣਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਏ) ਇਸਭਾਗਵਿਚਕਹਾਣੀਆਂਅਤੇਕਹਾਣੀਲੇਖਕਾਂਸੰਬੰਧੀਲਘੂਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚੋਂਕੁਲ 16ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਚਾਰਸਵਾਲਾਂਵਿਚੋਂ ਦੋ ਕਰਨੇਹੋਣਗੇ,ਜਿਸਵਿਚੋਂਹਰ ਸਵਾਲਪੰਜਅੰਕਾਂਦਾਹੋਵੇਗਾ।ਬਾਕੀਸੱਤਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂਪੰਜਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿ ਦੋ-ਦੋ ਅੰਕਦੇਹੋਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

2. ਯੂਨਿਟ ਦੋ ਵਿਚਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਤਨਿਬੰਧਲੇਖਣ, ਮੁਹਾਵਰਿਆਂਅਤੇਅਖਾਣਾਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁੱਲ 20 ਅੰਕਹੋਣਗੇ।

(ੳ) ਇਸਭਾਗਵਿਚਪੰਜਾਬੀਸਭਿਆਚਾਰਨਾਲਸੰਬੰਧਤਨਿਬੰਧਲਿਖਣਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਭ) ਇਸਭਾਗਵਿਚਮੁਹਾਵਰਿਆਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂਸੰਬੰਧੀਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

(ਏ) ਇਸਭਾਗਵਿਚਅਖਾਣਾਂਦੀਵਾਕਾਂਵਿਚਵਰਤੋਂਬਾਰੇਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।

ਨੋਟ: ਇਸਯੂਨਿਟਵਿਚਕੁੱਲ 16ਸਵਾਲਪੁੱਛੇਜਾਣਗੇ।ਪਹਿਲੇਦੋਸਵਾਲਾਂਵਿਚੋਂਇਕਸਵਾਲਹੱਲਕਰਨਾਹੋਵੇਗਾ ਜੋ ਦਸਅੰਕਾਂਦਾਹੋਵੇਗਾ।ਬਾਕੀ14 ਲਘੂਸਵਾਲਹੋਣਗੇ, ਜਿਨ੍ਹਾਂਵਿਚੋਂ10ਸਵਾਲਹੱਲਕਰਨੇਹੋਣਗੇ, ਜੋ ਕਿਇਕ-ਇਕਅੰਕਾਂਦੇਹੋਣਗੇ।ਇਸਯੂਨਿਟਦੇਕੁਲਅੰਕ 20 ਹੋਣਗੇ।

ਨਿਰਧਾਰਿਤਪੁਸਤਕਾਂ

1. ਕਥਾ-ਕਹਾਣੀ, ਡਾ. ਧਨਵੰਤਕੌਰ (ਸੰਪਾ.), ਪਬਲੀਕੇਸ਼ਨਬਿਊਰੋ, ਪੰਜਾਬਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ,2012
2. ਕਹਾਣੀਸ਼ਾਸਤਰ, ਸੰਤਸਿੰਘਸੇਖੋਂ, ਲੋਕਗੀਤਪ੍ਰਕਾਸ਼ਨ, ਸਰਹਿੰਦ, 1991
3. ਪੰਜਾਬੀਕਹਾਣੀਅਧਿਐਨ, ਟੀ.ਆਰ.ਵਿਨੋਦ, ਰਵੀਸਾਹਿਤਪ੍ਰਕਾਸ਼ਨ, ਅੰਮ੍ਰਿਤਸਰ, 1988
4. ਕਾਲਜਪੰਜਾਬੀਵਿਆਕਰਣ, ਡਾ.ਹਰਕੀਰਤਸਿੰਘਅਤੇਗਿਆਨੀਲਾਲਸਿੰਘ, ਪੰਜਾਬਸਟੇਟਯੂਨੀਵਰਸਿਟੀਟੈਕਸਟਬੁੱਕਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
5. ਲੋਕਆਖਦੇਹਨ, ਵਣਜਾਰਾਬੇਦੀ, ਆਰਸੀਪਬਲੀਕੇਸ਼ਨ, 2006
6. ਲੋਕਸਿਆਣਪਾਂ: ਅਖਾਣਤੇਮੁਹਾਵਰੇ, ਸੁਖਦੇਵਮਾਦਪੁਰੀ, ਲਾਹੌਰਬੁੱਕਸ਼ਾਪ, ਲੁਧਿਆਣਾ

STAT-M-111(Theory):STATISTICALMETHODS

Programme Outcome: The course aims to shape the attitudes of learners regarding the field of statistics**.**Theobjectiveofthecourseistomakethestudentsfamiliarwithvarioustechniquesused insummarization,presentationandanalysisofdata.Thefocuswillbebothontheoreticalaswell as practical approach

CourseOutcome: Oncompletionofthiscoursestudentswillbeableto:

(i) Getfamiliarwithvarioustechniquesusedinsummarizationandanalysisofdata.
(ii) Understandthenatureofstatisticaldata.
(iii) Understand about the types of correlations for bivariate and multivariatedata andtheir uses in real life problems.
(iv) Highlighttheconceptoflinearregressionanditsapplicationsinreallife
(v) Understandthevariousmeasuresofassociationbetweenattributes.

Class	**Sem.**	**CourseCode**	**CourseTitle**	**Credits=6**		
B.A./B.Sc.	**I**	**STAT-M-111**	**STATISTICALMETHODS**	**L=4**	**T=1**	**P=1**
TotalMarks(Theory=100)(Tutorial+Practical=50)InternalAssessment=20%						
TotalHours=105hours(Theory=60hours+Tutorial=15hours+Practical=30hours)						
TimeAllowedforExamination:Theory=3hours;Practical=3hours						
InstructionsforPapersetter	Therewillbeinallnine(9)questions,allofequalmarks. Thefirstquestionis**compulsory**andwillbeofshortanswer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set fromeachUnit					
InstructionsforCandidate	Thecandidatewillberequiredtoattemptfivequestionsinall includingthe compulsory first question and two questions fromeach Unit**.**					

CourseContents:

UNITI

Introduction:DefinitionandscopeofStatistics,conceptsofstatistical populationandsample.Scalesofmeasurement-nominal,ordinal, interval and ratio. Variables and attributes, Diagrammatical Representation of Data, Summarization of Data: Frequency Distribution and Graphical Presentation (histogram, frequency polygon and ogives). Measures of Central Tendency: mathematical and positional. Measures ofDispersion:range,quartiledeviation,meandeviation,standard deviation, coefficient of variation, moments, measures of skewness and kurtosis and Box- whisker Plot.

UNITII

Bivariate data: Definition, scatter diagram, simple correlation, rank correlation(with ties and without ties). Partial and Multiple Correlation Coefficients. Only definition and its applications (No derivation)

Fitting of linear and quadratic regression equations using principle of least squares.Theory of attributes and consistency of data, independence andassociationofattributes,measuresofassociationandcontingency for 2 x 2 and r x s contingency tables.

Notes:

1. **This course is not for the students of B.A./B.Sc. (General) Statistics Discipline Specific courses (Major/Core). Any other candidate shall be offered this minor subject in B.A./B.Sc.**
2. **Students are to be engaged to 7 hours of teaching per week(4lecturesfortheory,onelecturefortutorialand2 lectures for practical).**
3. **PracticalExaminationwillbeheldintherespective/samesemester.**

SUGGESTEDREADINGS:

1. Goon,A.M.,Gupta,M.K.andDasgupta,B.(2013):FundamentalofStatistics, Vol I, World Press,Kolkata
2. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2011): FundamentalsofStatistics,VolII, WorldPress,Kolkata.
3. Miller, IrwinandMiller,Marylees(2006):JohnE. Freund'sMathematical Statistics withApplications, (7th Edn.), Pearson Education,Asia.
4. Mood, A.M. Graybill, F.A. and Boes, D.C. (2011): Introduction to theTheoryofStatistics,3rdEdn.,(IndianEdition),TataMcGraw-HillPub. Co.Ltd.

STAT-M-112(PRACTICAL):STATISTICALMETHODS

(Credit:1)Practical:Max.Marks:25

PracticalwillbebasedonPaperSTAT-M-112(PRACTICAL):STATISTICALMETHODS

Practical: FinalWrittenExamination:15marks,InternalAssessment:5 marks, viva-voce: 5 marks

NoteforPractical: Thepracticalquestionpaperwillconsistof5questions. Thestudents will be required to attempt any three questions in three hours duration.